

**Access, Attitudes, and Need: A Behavioral Model Analysis of Mental Health and
Substance Use Treatment among Adults in the United States**

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Abstract

Despite the increasing prevalence of behavioral-health disorders in the United States, only a small fraction of adults in need receive professional treatment. Using data from the 2023 National Survey on Drug Use and Health (NSDUH), this study examined demographic, socioeconomic, clinical, and attitudinal predictors of past-year treatment for mental health and substance use problems among adults aged 18 and older. Guided by Andersen's Behavioral Model and the Theory of Planned Behavior, the analysis implemented a multi-model framework in which Group LASSO was used for structured variable selection, followed by survey-weighted generalized linear models and Random Forest classifiers to estimate and validate predictors. Across both treatment domains, need-based indicators and structural enabling factors were the most influential predictors of service use. Adults who experienced a major depressive episode, used illicit drugs, or reported psychological distress were substantially more likely to receive mental health treatment, whereas alcohol use disorder, marijuana involvement, and other substance-related indicators were the strongest predictors of substance use treatment. Age, race/ethnicity, and income showed consistent disparities, with older adults and racial/ethnic minority groups exhibiting lower utilization even after accounting for clinical need. Random Forest models corroborated these findings and highlighted nonlinear effects of distress, substance involvement, and socioeconomic resources, underscoring the complexity of treatment engagement. The results suggest that behavioral health access in the U.S. is shaped by the intersection of clinical severity, structural opportunity, and persistent social inequality. The overlap in predictors across both treatment types also emphasizes the need for more integrated and coordinated behavioral-health systems that can address co-occurring mental health and substance use conditions simultaneously.

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Behavioral-health disorders, which encompass both substance use and mental health issues, represent a major public health challenge that affects individuals, families, and communities across the United States (SAMHSA, 2023; SAMHSA, 2025). Despite the recognized need for care, national data consistently show that treatment engagement has not kept pace with the high prevalence of behavioral-health problems. In 2023, fewer than one in four adults aged 18 or older (23.0 percent) received mental health treatment in the past year, reflecting a persistent gap between need and utilization (SAMHSA, 2023; SAMHSA, 2025). This gap arises from a combination of structural and psychological barriers. Common structural obstacles include the cost of treatment (reported by 42.4 percent of adults with unmet SUD treatment need), limited insurance coverage (31.9 percent), and uncertainty about how or where to access services (38.7 percent). These external barriers intersect with powerful psychological factors, most notably the belief that one should be able to handle substance use independently, which is endorsed by 74.1 percent of adults with unmet SUD treatment need (SAMHSA, 2023; SAMHSA, 2025).

These patterns align closely with two behavioral-science frameworks that guide the present study. Andersen's Behavioral Model emphasizes how predisposing characteristics, enabling resources, and clinical need shape service utilization, while Ajzen's Theory of Planned Behavior highlights the roles of attitudes, subjective norms, and perceived behavioral control in shaping help-seeking intentions (Ajzen, 1985; Andersen, 1995; Andersen & Newman, 1973). These models offer a lens for understanding why high levels of clinical need do not routinely translate into treatment use. Understanding how structural and psychological barriers map onto

these theoretical pathways is essential for identifying which groups face the greatest obstacles and for guiding interventions that more effectively support engagement in behavioral-health care.

The present study directly addresses this ongoing challenge by utilizing nationally representative data to examine who accesses behavioral-health treatment and under what circumstances. Specifically, this research uses data drawn from the 2023 National Survey on Drug Use and Health (NSDUH) to focus on two distinct yet interrelated outcomes: the receipt of mental health treatment and the receipt of substance use treatment within the past year. The extensive information collected by the NSDUH details treatment use, allowing for the analysis of factors such as demographic characteristics (e.g., age, race/ethnicity), clinical status (e.g., severity of SUD or presence of mental illness), and attitudinal factors (e.g., perceived need for treatment and reasons for not receiving care). By analyzing these outcomes within a unified framework, this research seeks to clarify how these numerous factors jointly influence treatment access, thereby providing critical information to assist researchers, clinicians, and policymakers to better understand and improve the nation's behavioral health. The study not only identifies key predictors of service use, but also situates them within well-established behavioral-science theories of help-seeking, specifically Andersen's Behavioral Model and the Theory of Planned Behavior, thereby providing a robust theoretical context for interpreting complex utilization patterns and informing efforts to improve the nation's behavioral health.

Literature Review

Theoretical Foundations

The present study is grounded in two complementary frameworks that together explain how individuals recognize, evaluate, and act upon their behavioral-health needs: Andersen's

Behavioral Model of Health Service Use and Ajzen's Theory of Planned Behavior (TPB) (Ajzen, 1985; Andersen, 1995; Andersen & Newman, 1973). Each addresses a distinct but interrelated dimension of treatment access with one emphasizing structural opportunity and equity, and the other focusing on psychological motivation and intention. Integrating both allows for a more comprehensive understanding of why some individuals seek mental health or substance use treatment while others do not, even under comparable conditions of need.

Andersen Behavioral Model

Originally developed by Andersen and Newman (1973), the Behavioral Model conceptualizes health-service utilization as a function of three domains: predisposing characteristics, enabling resources, and need-based factors. Andersen's later revision (1995) expanded the model by explicitly incorporating environmental and health-system contexts, recognizing that utilization patterns are shaped not only by individual characteristics but also by broader social structures. Subsequent adaptations have further extended the model into a dynamic framework. For example, the Gelberg-Andersen Behavioral Model for Vulnerable Populations introduced additional domains relevant to people experiencing homelessness, poverty, or social marginalization, emphasizing how competing survival needs and structural barriers alter traditional pathways to care (Gelberg et al., 2000). More recent work conceptualizes the model as recursive rather than linear, acknowledging that experiences with treatment whether successful, ineffective, or stigmatizing feed back into future perceptions of need, expectations of efficacy, and willingness to reengage with services (Arcury et al., 2012; Kadden and Litt, 2011). These developments illustrate how the Behavioral Model has evolved from a static sociological classification into a flexible, multilevel framework that links structural context with psychological and experiential processes.

Beyond its sociological roots, the Behavioral Model has evolved into a dynamic framework that incorporates feedback loops between health outcomes and future help-seeking behavior. Experiences with treatment, either positive or negative, would shape subsequent perceptions of need, expectations of efficacy, and willingness to reengage with care. This recursive structure aligns closely with psychological models of intention and behavior, suggesting that service use is not merely a function of access, but also of belief, motivation, and perceived control. A substantial body of empirical work has provided support for these relationships proposed by Andersen's model. Choi et al. (2014) demonstrated that predisposing, enabling, and need factors each contribute to treatment utilization, showing, for example, that adults over 65 were significantly less likely to use behavioral-health services even after accounting for clinical need, while greater symptom severity strongly increased treatment odds. Similar patterns were found in Romo et al. (2018), whose application of the Gelberg–Andersen model to adults with opioid use disorder revealed that predisposing risks (e.g., recent arrest), enabling resources (e.g., insurance and geographic location), and need severity (e.g., dependence vs. abuse) each independently predicted specialty treatment receipt. Wu et al. (2017) observed the same structure among adults with cannabis use disorder: need indicators such as comorbid mental health or other substance use disorders increased treatment utilization, while certain predisposing and enabling characteristics such as gender, education, and race/ethnicity reduced it. Together, these studies provide strong empirical evidence that treatment engagement is shaped by the interplay of predisposing characteristics, enabling resources, and clinical need, consistent with the core assumptions of Andersen's Behavioral Model.

Theory of Planned Behavior (TPB)

Ajzen's Theory of Planned Behavior (1985) extends this psychological dimension by explaining how individuals form the intention to act. TPB posits that behavior is guided by three cognitive determinants: attitudes, subjective norms, and perceived behavioral control. Attitudes represent a person's evaluation of whether seeking treatment is beneficial or necessary; subjective norms reflect perceived social expectations regarding help-seeking; and perceived behavioral control captures the individual's belief in their ability to obtain treatment given available resources and obstacles.

In behavioral-health contexts, these constructs map directly onto common barriers and facilitators of treatment. Positive attitudes toward care and strong social support predict higher help-seeking, while stigma and unsupportive norms deter it (Adams et al., 2022; Liddon et al., 2017; Walker et al., 2015). Perceived control, which is closely tied to enabling factors in Andersen's model, such as insurance or financial resources, determines whether intention translates into action. Individuals who view treatment as valuable, socially accepted, and attainable are far more likely to pursue it, even when facing clinical distress or social stigma.

A growing body of research also provides strong empirical support for the relationships proposed by the TPB in explaining mental health and substance use help-seeking. In a scoping review of 39 studies, Adams et al. (2022) found that attitudes toward services and perceived behavioral control were the most consistent predictors of help-seeking intention, while subjective norms also played a significant, though somewhat less universal, role; moreover, both intentions and perceived behavioral control predicted actual help-seeking behavior in the majority of studies that measured behavioral outcomes. More targeted work by Chandrasekara (2016) similarly demonstrated that positive attitudes and perceived control significantly increased

intention to seek psychological services among university students, with stigma, reflecting unsupportive social norms, emerging as a prominent barrier even when norms did not directly predict intention. Research on substance use treatment echoes this pattern: Stringer and Baker (2015) showed that stigma rooted in gendered and parental expectations operates as a powerful normative deterrent, with women more likely than men to cite stigma as a barrier to care. Collectively, these studies provide robust evidence that attitudes, subjective norms, and perceived behavioral control jointly shape treatment-seeking intentions and behavior in ways that align closely with Ajzen's theoretical model.

By integrating Andersen's and Ajzen's frameworks, this study links structural conditions with psychological processes, offering a multilevel explanation of behavioral-health treatment access. Andersen's model situates individuals within a broader context of opportunity and constraint, while TPB captures the cognitive mechanisms that mediate between perceived need and actual behavior. Together, they provide the theoretical foundation for the current analysis: one that recognizes behavioral-health service utilization as the product of both social structure and human agency: a reflection of what people are able, expected, and motivated to do in response to mental health and substance use challenges.

Demographic Predictors

Demographic factors consistently represent some of the strongest and most stable predictors of behavioral-health treatment utilization. Within Andersen's Behavioral Model, these characteristics are classified as predisposing factors, shaping an individual's overall propensity to seek care before any actual need or structural opportunity comes into play. In both mental health and substance use contexts, gender, race, ethnicity, and age emerge repeatedly as significant correlates of treatment use, though the direction and magnitude of these effects vary depending

on population and condition. Importantly, these demographic patterns do not operate in isolation. Within Andersen's Behavioral Model, they shape predisposing tendencies and differential enabling resources, while in Ajzen's Theory of Planned Behavior they influence how people interpret symptoms, experience stigma, and judge whether seeking treatment is socially acceptable or personally manageable. In other words, demographic disparities often reflect deeper psychological and structural mechanisms.

A large body of research has shown that gender plays a central role in predicting treatment access, with women are generally more likely to seek mental health treatment than men, often due to greater emotional awareness and lower stigma toward discussing distress (Liddon et al., 2018; Owen, 2013). For instance, Owens (2013) found that across health services, women were more frequent users than men, yet they still reported gaps in counseling quality and provider communication. Liddon et al. (2018) similarly observed that young adult women reported higher willingness to engage with mental health professionals, driven by higher self-reported perceived need and greater openness to vulnerability. However, these trends shift somewhat in substance use contexts: studies such as Chen et al. (2013) and Gilbert et al. (2019) show that although men are more likely to develop substance use disorders and enter treatment, women face stronger attitudinal barriers and stigma that often prevent them from initiating care. Many women report wanting to "handle it themselves" or believe that problems would resolve without intervention (Gilbert et al., 2019). In short, while women display higher mental health service utilization, men tend to dominate in substance use treatment entry, reflecting how gender norms interact differently across these two forms of behavioral-health care (Chen et al., 2013; Gilbert et al., 2019; Liddon et al., 2017; Stringer & Baker, 2015; Sullivan, 2022; Walker et al., 2015). These gendered norms map directly onto TPB pathways: women tend to report higher

perceived need and fewer normative barriers for mental health care, whereas men often endorse norms of self-reliance that lower perceived behavioral control and intention to seek help, especially for substance use.

Race and ethnicity represent another consistent axis of disparity across both treatment domains. Numerous studies have shown that Black and Hispanic individuals are less likely to initiate or complete treatment compared to White counterparts, even after accounting for socioeconomic status and disorder severity: for example, Chen and Rizzo (2010) reported that Latino and Black adults used fewer psychotherapy services and spent less on mental health care compared to Whites, largely due to language barriers and insurance coverage gaps. In adolescent populations, Cummings et al. (2011) found that Black and Hispanic youth were significantly less likely to receive substance use treatment than White adolescents, despite similar levels of reported need. These disparities persist throughout adulthood and even into older age groups. Acevedo et al. (2012) noted that Black clients were less likely than White or Native American clients to initiate substance use treatment, though engagement rates, once treatment began, were similar. However, Guerrero (2013) found that treatment completion rates were still lower among African American and Latino clients, suggesting that systemic barriers persist throughout the continuum of care. From a theoretical standpoint, these disparities illustrate how predisposing characteristics interact with enabling resources, such as insurance, language access, and culturally responsive services, and how subjective norms within communities shape attitudes toward formal treatment.

Recent work has also examined how racial/ethnic disparities evolve across the lifespan. Using a nationally representative dataset, Vasilenko (2017) demonstrated that racial and gender differences in substance use disorders vary substantially with age. In younger adulthood, White

individuals had the highest rates of alcohol and drug disorders, but these disparities reversed later in life, with Black adults exhibiting higher prevalence during midlife. This pattern, sometimes referred to as the “crossover effect,” suggests that early life social and structural advantages among White individuals may diminish over time, while cumulative stressors among Black adults compound risk and potentially lower treatment engagement later in life. These findings align with Andersen’s model, which recognizes that demographic influences do not operate in isolation but intersect dynamically with social and environmental contexts. Age-graded changes in social norms and perceived control also fit TPB predictions: younger adults often endorse more open attitudes toward therapy but lower structural stability, while older adults may have greater resources but weaker perceived need or stronger generational stigma.

Research on age differences similarly indicates that behavioral-health service utilization varies across the life course. Older adults are consistently less likely to perceive a need for, or seek, behavioral-health treatment compared to younger adults. Choi et al. (2014) found that older adults were less likely to access either mental health or substance use treatment and often cited cost and lack of readiness as major barriers. Interestingly, stigma was reported less often by older adults, possibly reflecting generational norms that normalize distress as part of aging rather than as a treatable problem. Cuijpers et al. (2020) added that the effectiveness of psychotherapy for depression tends to be lower among youth and consistent across middle-aged and older adults, suggesting that developmental stage also shapes not only treatment entry but treatment outcomes.

Although gender, race, and age have been the most thoroughly studied demographic predictors, their combined or intersectional effects are increasingly recognized as critical for understanding behavioral-health access. For instance, Lauffenburger et al. (2013) observed that

Black and Hispanic women had the lowest adherence to evidence-based mental health treatments, even after controlling for socioeconomic and clinical factors. Similarly, Wintersteen et al. (2005) found that matching clients and therapists by gender and race improved engagement and retention, particularly among female and minority adolescents. These findings reinforce that treatment experiences are shaped by more than just individual need: they are filtered through overlapping social identities that influence comfort, trust, and perceived cultural congruence in care. These patterns underscore Andersen's view that predisposing characteristics intersect with social context, while TPB helps explain why overlapping identities shape perceived legitimacy, comfort, and control over seeking treatment.

A similar intersectional perspective has emerged in substance use research. Windsor et al. (2014) conducted a meta-analysis showing that cognitive-behavioral therapy (CBT), a common intervention for substance use disorders, was less effective for Black and Hispanic clients than for White clients, largely because traditional CBT frameworks often overlook culturally specific values, stressors, and community contexts. Likewise, Pinedo (2019) emphasized that cultural stigma and fear of discrimination discourage Latino individuals from seeking formal treatment, leading many to rely on informal or faith-based support networks instead. These findings underscore that racial and cultural background not only affect whether treatment is sought, but also how effective and sustainable that treatment becomes once accessed.

These studies paint a consistent picture: demographic factors are powerful but complex predictors of behavioral-health treatment utilization. Women and racial minorities often face more attitudinal and structural barriers, while men and White individuals show higher raw participation in certain types of services but may receive less benefit from them. Age adds another layer, shaping both perceived need and type of service sought. From Andersen's

perspective, these variables reflect predisposing characteristics that interact with enabling and need-based factors to shape help-seeking patterns. In simpler terms, who people are and how society perceives them still plays a decisive role in whether they receive behavioral-health care, despite decades of policy efforts aimed at achieving equity in access.

Socioeconomic and Structural Predictors

Socioeconomic status and structural conditions play an essential role in determining whether people are actually able to access behavioral-health treatment when they need it. Within Andersen's Behavioral Model, these variables are described as enabling factors, meaning they are the tangible and social resources that make it either possible or impossible for someone to seek help. Although clinical need and personal motivation matter a great deal, socioeconomic resources such as income, education, and insurance coverage shape the context in which those needs are recognized and acted upon.

Among these, health insurance has consistently emerged as one of the strongest predictors of behavioral-health service use. Studies have repeatedly shown that people with any form of coverage, especially private insurance or Medicaid, are far more likely to receive both mental health and substance use treatment than those who are uninsured. Walker et al. (2015) found that uninsured adults reported significantly lower treatment rates and much higher unmet need compared to insured adults, emphasizing that insurance provides the first gate to entry for most formal services. Similarly, Ali et al. (2015) showed that individuals with Medicaid or private insurance were more likely to receive substance use treatment, but only when they actually perceived a need for it. In other words, coverage expands opportunity, but it doesn't necessarily translate into use if people don't believe they need help or don't view available services as acceptable.

Insurance alone also fails to eliminate disparities once people enter care. Even among insured adults, nearly half in Walker et al.'s (2015) study still reported attitudinal barriers such as wanting to “handle it themselves,” revealing that structural access does not automatically remove psychological resistance. Benedict (2006) reached a similar conclusion in a national survey of children with functional limitations, showing that public insurance and higher parental education predicted greater service use but also higher reports of unmet needs, perhaps because awareness and expectations rise alongside access.

Employment and income are another major dimension of structural access. Driscoll and Bernstein (2012) compared employed and unemployed adults and found that unemployment was linked to worse health and substantially lower likelihood of receiving needed care, mainly due to loss of employer-sponsored insurance and the inability to afford services. Unemployed adults were more likely to delay or skip care entirely, even when they had some form of coverage, because out-of-pocket costs and copayments were unaffordable. These findings underline that employment provides more than income, it also supplies stability and access to benefits that shape long-term engagement with the health system.

Education and health literacy also shape access in important ways. Education provides not only information about symptoms and treatment options but also the skills needed to navigate complex healthcare systems. Sharland et al. (2023) found that individuals without formal educational qualifications were consistently underrepresented in England's national psychological-therapy program, even after controlling for disorder severity. Low education may reduce awareness of available services or reinforce stigma about seeking help. Conversely, Benedict (2006) found that families with higher education were more likely to advocate for care

and to recognize unmet needs, implying that awareness can raise expectations for quality and adequacy of treatment.

At the same time, structural factors extend beyond financial and informational resources. Stigma and life circumstances often intersect with these barriers in ways that disproportionately affect women and parents. Stringer and Baker (2015) highlighted that many treatment programs lack childcare support, transportation options, and flexible scheduling: practical issues that make attendance difficult, especially for mothers. These barriers illustrate that access is not just about affordability; it also depends on whether the structure of the service system accommodates real-world responsibilities and social expectations.

Overall, the evidence suggests that socioeconomic and structural variables operate through both opportunity and constraint. Insurance coverage, education, and employment expand the potential for treatment utilization by reducing cost barriers and increasing awareness. Yet, even when these resources are present, inequities persist because of deeper systemic and social barriers: cost sharing, inflexible services, or stigma embedded within communities and care systems. From the perspective of Andersen's Behavioral Model, these findings reaffirm that enabling factors mediate the link between need and actual service use. Behavioral-health access, therefore, depends not only on recognizing that help is needed but also on whether one's environment and circumstances allow that recognition to translate into action.

Behavioral and Clinical Predictors

Beyond demographic and socioeconomic influences, a large body of research has shown that behavioral and clinical characteristics are strong correlates of whether people ultimately seek mental health or substance use treatment. In Andersen's Behavioral Model, these factors are generally classified as need-related, representing both the perceived need and the evaluated

clinical need for care. In other words, while predisposing and enabling factors determine who can access treatment, behavioral and clinical conditions often determine who feels they must.

Early work by Harris and Edlund (2005) demonstrated that the presence of a diagnosable mental disorder dramatically increases the likelihood of treatment utilization. Using national survey data, they found that adults who met diagnostic criteria for a mental disorder were several times more likely to use outpatient services than those without such disorders, particularly when symptoms caused functional impairment or distress. However, perceived need was a crucial mediator, with many respondents with clinically significant symptoms did not view themselves as needing treatment, a pattern that remains consistent across later research.

In substance use contexts, the role of clinical need and behavioral comorbidity becomes even more complex. Hser et al. (1998) examined longitudinal treatment outcomes among individuals with SUDs and found that those with co-occurring mental disorders or more severe patterns of substance use were more likely to re-enter treatment over time. In essence, repeated cycles of relapse and remission often correspond to greater treatment contact, but not necessarily better recovery outcomes. Brown et al. (1999) highlighted a similar pattern in their study of patients with comorbid post-traumatic stress disorder (PTSD) and SUD. Patients with both PTSD and SUD used addiction treatment services more frequently, especially inpatient care, but were no more likely to receive adequate psychiatric treatment for PTSD itself. This mismatch between psychiatric and addiction services shows how overlapping behavioral conditions can lead to fragmented care, where one problem drives treatment entry but another remains untreated.

More recent large-scale epidemiologic studies have further clarified which clinical and behavioral characteristics predict treatment engagement. Blanco et al. (2015) analyzed data from

the National Epidemiologic Survey on Alcohol and Related Conditions (NESARC) and found that treatment-seeking probabilities vary substantially by disorder type. Individuals with drug dependence had the highest lifetime probability of seeking treatment (up to 90%), followed by those with alcohol dependence (54%) and drug abuse (60%). By contrast, individuals with alcohol abuse, generally a less severe form, showed extremely low treatment rates, at only 16%. These findings illustrate that treatment utilization increases with disorder severity but remains strongly affected by personal and contextual variables such as early age of onset, education, and marital status. Notably, higher education and older age cohorts were associated with lower probabilities of treatment, perhaps reflecting social stigma or self-reliance tendencies that reduce help-seeking despite clinical need.

Comorbidity with other psychiatric or medical conditions consistently emerges as one of the strongest behavioral predictors of treatment use, but it also highlights where clinical need fails to translate into actual service engagement. For instance, Wu et al. (2017), analyzing nine years of NSDUH data, found that adults with cannabis-use disorder frequently presented with additional conditions such as nicotine dependence, alcohol use disorder, or major depressive episodes, yet treatment rates remained extremely low of under 13%. Notably, even groups typically associated with higher resources, such as women, married adults, and college-educated individuals, showed especially low utilization despite significant impairment. This pattern underscores a recurring theme in the broader literature: clinical need rarely operates alone; instead, it is filtered through attitudes, stigma, and social conditions that shape whether people recognize or act upon their symptoms.

A related but more recent direction in the literature examines how behavioral and clinical need can be formalized and predicted through computational modeling. Chekroud et al. (2016)

demonstrated that complex behavioral data, such as sleep disturbance patterns, self-reported restlessness, and baseline symptom severity, can be used to anticipate treatment outcomes with reasonable accuracy. Although their study focused on antidepressant remission rather than treatment access, the broader implication aligns with Andersen's model: indicators of need are not merely diagnostic labels but patterned behavioral signals that predict whether individuals improve, persist, or disengage. Together with studies like Wu et al. (2017), this work suggests that understanding treatment utilization requires attending not only to clinical severity but also to the behavioral "signature" through which individuals experience and interpret their symptoms.

Clinical comorbidities also shape outcomes after treatment initiation. Weaver et al. (2003) and Romo et al. (2018) both found that individuals with co-occurring depression and substance use disorders had poorer retention and lower treatment satisfaction compared to those with a single diagnosis. These studies emphasize the "dual-diagnosis" challenge, where individuals with both mental and substance use disorders require more integrated care models but often encounter systems that treat these problems separately. Sullivan (2022) similarly argued that fragmented treatment systems contribute to persistent disparities in behavioral-health recovery, since the severity of one disorder may mask or complicate recognition of another.

Taken together, they illustrate that behavioral and clinical predictors operate primarily through need recognition, disorder severity, and comorbidity. People who experience greater functional impairment, more intense symptoms, or multiple overlapping disorders are statistically more likely to enter treatment, but this relationship is far from linear. As Wu et al. (2017) and Blanco et al. (2015) both note, high prevalence of comorbidity does not guarantee proportional treatment engagement. Instead, the translation from need to action depends on

individual insight, stigma, and systemic fragmentation. Within Andersen's framework, these findings confirm that clinical need alone cannot explain utilization; it interacts continuously with predisposing and enabling factors to shape real-world patterns of care.

Attitudinal and Risk-Perception Factors

Even when people have both the resources and the clinical need for care, a large portion of help-seeking behavior is shaped by internal beliefs about treatment—what people think will happen, what they believe others expect of them, and whether they feel capable of pursuing care. These attitudinal processes map closely onto Ajzen's Theory of Planned Behavior (1985), which emphasizes attitudes, subjective norms, and perceived behavioral control as core determinants of behavioral intention. Within behavioral health, these psychological factors often explain why two individuals with similar symptoms and identical access conditions still differ dramatically in whether they seek treatment.

Attitudes and Perceived Behavioral Control

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Risk Perception and Perceived Need

Perceptions of risk also shape whether people feel that treatment is necessary. National trend data consistently show that individuals who view substance use as risky are less likely to use substances and more likely to support intervention. Lipari's (2013) analysis of NSDUH reports showed clear gradients in substance use based on perceived harm, especially among adolescents, and similar patterns appear across adults when perceived treatment need is considered. Askari et al. (2021), for example, found that despite rising impairment associated with cannabis-use disorder, only a small fraction of adults perceived a need for treatment. In both cases, perceived risk acts as a cognitive filter: if substance use does not feel dangerous or consequential, treatment rarely feels urgent.

Social Norms and Cultural Influences

Social norms also exert a powerful influence on help-seeking, especially when treatment carries social meaning. In substance use contexts, peer and cultural perceptions often normalize behavior; veterans in Wilkinson et al. (2016), for instance, described cannabis as natural and non-addictive, framing use as socially acceptable despite institutional messaging to the contrary. In mental health contexts, stigma remains a major deterrent. Hughes et al. (2016) showed that perceived public stigma reduces willingness to disclose symptoms or engage in therapy, particularly in masculine or collectivist cultures. These normative pressures lower behavioral intentions even when attitudes toward treatment are moderately positive, and they help explain persistent underutilization in groups where distress is viewed as personal weakness or something to be managed privately.

Protective Attitudes and Cognitive Beliefs

At the same time, not all attitudinal influences are barriers: some beliefs act as protective factors that prevent escalation or increase resilience. Traits such as optimism, mindfulness, and anti-drug beliefs, synthesized in Nawi et al.'s (2021) review, appear to buffer adolescents from initiating substance use in the first place. Similar cognitive strengths, especially self-efficacy, continue to matter throughout treatment engagement. Kadden and Litt (2011) describe a feedback loop where individuals who successfully manage small steps in recovery gradually strengthen their sense of control, which in turn increases the likelihood of sustained engagement. These protective attitudes reflect the positive end of Ajzen's constructs and show how motivation can be strengthened rather than merely assessed.

Changing Contexts and Medical Legitimacy

Public attitudes also shift in response to larger societal changes. The growing medicalization and legalization of marijuana illustrate how evolving social norms can reshape perceptions of harm and legitimacy. Azcarate (2020) found that many adults who use cannabis for medical reasons choose not to disclose this information to clinicians, often because they anticipate disapproval or dismissiveness. This mismatch between personal beliefs and perceived professional norms highlights how trust, legitimacy, and expectations about providers shape whether people engage openly with formal treatment systems.

Across these themes, a clear pattern emerges: attitudinal and perceptual variables shape not only whether individuals intend to seek help but whether they follow through once opportunities exist. From a TPB perspective, attitudes (expected outcomes), subjective norms (social approval), and perceived control (self-efficacy and logistical access) interact continuously to form intentions. When mapped back onto Andersen's Behavioral Model, these

same constructs function as predisposing characteristics that influence how enabling resources and clinical need translate into action. In practical terms, even when treatment is available and affordable, many individuals do not pursue it if they perceive low personal risk, distrust professional systems, anticipate stigma, or doubt their ability to benefit. This literature makes clear that expanding treatment access requires more than structural change—it requires changing the beliefs, norms, and motivational frameworks that guide help-seeking.

The Present Study

Across decades of research, behavioral-health service use has been shown to depend on a dynamic interplay of personal, social, and structural factors. Andersen's Behavioral Model emphasizes how predisposing characteristics, enabling resources, and clinical need shape service utilization, while Ajzen's Theory of Planned Behavior highlights the cognitive processes (attitudes, norms, and perceived control) that influence whether individuals act on their needs. Together, these frameworks suggest that treatment use reflects not only symptom severity or resource availability, but also how individuals interpret, value, and respond to their behavioral-health experiences within their social environments.

Although previous research has documented persistent disparities by gender, race, and socioeconomic status, age-related differences in treatment utilization remain less systematically understood, particularly among adults. Existing studies provide mixed evidence: some suggest that older adults underutilize care due to stigma, low perceived need, and limited access, whereas others find age differences diminish after adjusting for clinical and socioeconomic factors. This inconsistency reveals a notable gap. Age is often treated as a background covariate rather than a theoretically meaningful predictor, despite population aging and rising behavioral-health needs

among U.S. adults. Understanding how age interacts with structural, clinical, and attitudinal factors is crucial for clarifying where gaps in care emerge across the adult life span.

The present study seeks to address this gap by identifying and comparing the strongest predictors of mental health and substance use treatment utilization among U.S. adults using data from the 2023 National Survey on Drug Use and Health (NSDUH). To do so, the study employs a multi-model analytic strategy that integrates group LASSO with size-normalized penalties, survey-weighted logistic regression, and Random Forest classifiers. These methods offer complementary strengths: group LASSO provides a principled way to identify the most influential predictor domains; survey-weighted logistic regression generates population-representative, interpretable odds ratios; and Random Forest models capture nonlinear or interaction patterns that may explain why certain demographic groups engage differently with behavioral-health services. This approach also allows the study to evaluate whether the predictors of mental health treatment mirror or diverge from those of substance use treatment. Prior work often examines these domains separately, but emerging research, including the present findings, suggests that many clinical indicators, such as major depressive episodes, alcohol involvement, and illicit drug use, operate across both systems. Comparing models side by side therefore offers insight into whether behavioral-health treatment in the U.S. functions as two distinct systems or as parallel components of a broader, interconnected care continuum.

By centering age alongside well-established predictors such as gender, race/ethnicity, insurance coverage, and income, the study deepens existing theoretical models of treatment-seeking. Andersen's framework helps structure the roles of resources and need, while Ajzen's framework informs the motivational and attitudinal pathways that may facilitate, or inhibit help-seeking among different adult age groups. Ultimately, this design moves beyond

descriptive comparisons and evaluates how predisposing, enabling, need-based, and attitudinal factors jointly influence who receives behavioral-health care in the United States.

Hypotheses:

Guided by Andersen's Behavioral Model and the Theory of Planned Behavior, we formulated three hypotheses:

1. Structural enablers, specifically insurance coverage, income, and education, and indicators of clinical need (e.g., depression, distress, and substance use severity) would be the strongest positive predictors of treatment utilization.
2. Demographic and attitudinal variables, particularly gender, race/ethnicity, and perceived risk, would moderate these relationships, with underutilization persisting among racial/ethnic minorities, men, and individuals reporting low perceived risk of harm.
3. Age would emerge as a distinct and understudied factor influencing access, with older adults showing lower treatment utilization after controlling for other predictors.
4. Predictor overlap would be substantial across both mental health and substance use treatment, with many of the same structural, demographic, and clinical factors shaping engagement in both service systems.

Methods

Participants

The 2023 National Survey on Drug Use and Health (NSDUH) samples 56,705 U.S. residents aged 12 and older, but the present study restricted analyses to adults aged 18 and above, yielding an analytic sample of 45,133 adults with valid sampling weights. All models

incorporated the NSDUH respondent-level weight to ensure population-representative estimates under the survey's complex multistage design.

This adult-only restriction was necessary for both conceptual and methodological reasons. Several key predictors central to the study (including psychological distress (K6), employment status, and certain substance-use indicators) are measured differently or not at all among adolescents. Using a mixed-age sample would therefore introduce inconsistencies in measurement and limit comparability across predictors. Focusing on adults produces a cleaner, more internally valid model of treatment utilization. At the same time, this choice narrows the scope of the findings. Thus, the findings pertain specifically to adult pathways into mental health and substance use treatment.

Materials

This study used the 2023 National Survey on Drug Use and Health (NSDUH) Public Use File (PUF) (SAMHSA, 2023; SAMHSA 2025). The data is a nationally representative, cross-sectional survey administered annually by the Substance Abuse and Mental Health Services Administration (SAMHSA). The study focused exclusively on predictive modeling of treatment access, examining which demographic, socioeconomic, and clinical factors predict whether individuals receive mental health or substance use treatment in the past year. Given the cross-sectional nature of NSDUH data, all analyses were explicitly associative, and non-causal due to the lack of experimental manipulation. Nevertheless, these analyses inform hypotheses for future longitudinal or experimental studies. The analytic goal was to identify which factors best predict treatment access, thereby highlighting groups that are most at risk for unmet treatment needs.

Measures

The 2023 NSDUH data is subject to SAMHSA’s use policy. Permission to use the dataset is granted for research purposes under SAMHSA’s terms, and all analyses complied with these requirements. Inclusion criteria required complete data for key outcomes of interest, including treatment access, telehealth utilization, or stigma-related items, as well as valid survey weights. To handle missing data for critical outcomes and demographic information, this study utilized “imputation revised” (IR) variables, which had their missing values replaced with statistical imputation, and eliminated the need for respondent exclusion from analysis. Variables were selected based on their direct relevance to the study’s overarching objectives: predicting treatment access, examining telehealth adoption, and understanding the influence of stigma on help-seeking behaviors. Inclusion criteria emphasized variables that either represent primary outcomes of interest, capture key predictors of access and use, or provide critical demographic and socioeconomic context for interpreting disparities. Survey questions, detailed tables of variable types and description (Table A1-A2) can be found in Appendix A.

Treatment Access Variables

The primary outcomes were past-year receipt of substance use treatment (SUTRTPY) and past-year receipt of mental health treatment (MHTRTPY). These binary indicators directly operationalized the central research question: who is and is not accessing care?

Demographic Variables

Demographic characteristics were essential for understanding disparities in treatment access and digital health uptake. Age (CATAG6), sex assigned at birth (IRSEX), and race/ethnicity (NEWRACE2) captured well-documented demographic differences in health care access.

Socioeconomic Variables

Marital status (IRMARIT), highest level of education completed (IREDUHIGHST2), and employment status (IRWRKSTAT) reflected social determinants of health that often shape treatment-seeking behavior. Health insurance coverage (IRINSUR4) was a direct structural barrier or facilitator of treatment access, while family income (INCOME) measured economic resources strongly associated with both utilization and unmet need.

Behavioral and Clinical Variables

Predictors of treatment access included clinically significant conditions: past-year alcohol use disorder (IRPYUD5ALC), past-year marijuana use disorder (IRPYUD5MRJ), past-year major depressive episodes (IRAMDEYR), and serious psychological distress (KSSLR6YR). These were included because severity of substance use and mental health symptoms were among the strongest drivers of treatment seeking in epidemiological research.

Attitudinal and Risk Perception Variables

To capture broader risk profiles, the study included frequency of alcohol use (IRALCFY) and marijuana use (IRMJFY) as behavioral indicators of substance involvement. Perceptions of risk: risk of smoking cigarettes daily (RSKCIGPKD) and risk of using marijuana weekly (RSKMRJWK) reflected attitudinal barriers to treatment and prevention. Any illicit drug use in the past year (ILLYR) provided a general indicator of exposure to risk behaviors. These variables helped contextualize how patterns of use and perceptions of risk affect treatment engagement and perceptions of risk affect treatment engagement.

Design & Procedure

All analyses were conducted using *Jupyter Notebook* version 7.0.8 with the programming language *Python* version 3.10.7, utilizing packages: *pandas*, *numpy*, *matplotlib*, *statsmodel*, *group_lasso*, *imblearn* and *scikit-learn* for data processing, data analysis, and data visualization (Harris et al., 2020; Hunter, 2007; Kluyver et al., 2016; Lemaître et al., 2017; McKinney, 2010; Moe, 2019; Pedregosa et al., 2011; Seabold & Perktold, 2010; The pandas development team, 2025; Van Rossum & De Boer, 1991). To properly account for the 2023 National Survey on Drug Use and Health (NSDUH) complex multistage sampling design, all statistical modeling and inference were conducted using survey-specific procedures. Analyses incorporated the final person-level analysis weight (ANALWT2_C) to ensure population-level representativeness.

All categorical predictors were transformed using one-hot encoding, but no reference category was dropped during the penalized modeling stage. This choice was necessary because Group LASSO requires a complete set of dummy indicators for each categorical variable so that the entire group of coefficients can be penalized jointly. Retaining all levels ensures that the group index used in the penalty correctly maps each dummy variable to its parent predictor and avoids inconsistencies in the group structure (Simon et al., 2013). Although this produces k dummy variables for a k -category factor during the selection step, it does not affect interpretability, because the penalized model is used only for variable screening, not for reporting effect sizes (Simon et al., 2013).

For the survey-weighted GLM refit, predictors were recoded using the standard one-hot scheme with a dropped reference category, yielding interpretable contrasts of the form “category vs. reference.” This two-stage encoding strategy ensures that (1) the Group LASSO step

performs valid group-wise selection without distortion from dropped levels, and (2) the final logistic regression is interpretable and consistent with conventional practice. For example, the youngest age band served as the reference for age; “male” for sex at birth; “non-Hispanic White” for race/ethnicity; the lowest income bracket for income; “married” for marital status; and “uninsured” for insurance coverage where applicable.

Binary indicators (e.g., illicit drug use, alcohol- or marijuana-use disorder, insurance coverage) were retained in 0/1 form throughout all stages of analysis. Continuous predictors were standardized only inside the penalized fitting step to ensure that the Group LASSO penalty acted fairly across variables. In the final GLM refit, continuous variables were restored to their original scales to allow interpretation of odds ratios in familiar units. This two-step approach preserves proper penalization during selection while ensuring that the final coefficients reflect unbiased population-level contrasts under the complex survey design.

The primary analytic approach was a logistic regression model predicting the probability of treatment access from demographic, socioeconomic, behavioral, and clinical predictors. Logistic regression provides interpretable odds ratios and well-established inferential properties (Hosmer et al., 2013). To reduce dimensionality and guard against overfitting, the analysis first applied Group LASSO logistic regression as a variable-selection step. Group LASSO extends the L1 penalty to grouped parameters (such as the full set of dummy indicators for a categorical variable) allowing entire variables (rather than isolated levels) to be selected or excluded (Simon et al., 2013). This makes the approach well suited for high-dimensional survey data containing many nominal predictors, multilevel categorical structures, and sparse outcomes (Iparraguirre et al., 2023; Signorino & Kirchner, 2018; Tibshirani, 1996).

A key methodological challenge is that standard penalized regression procedures do not accommodate complex survey designs by default. Traditional cross-validation splits units at random, ignoring sampling weights and strata, which can underestimate variability and induce “design leakage” (Iparragirre et al., 2023). Although weighted LASSO formulations exist for survey estimation, they are primarily intended for variance reduction rather than for inference on penalized coefficients (Jasiak & Tuvaandorj, 2023; McConville et al., 2017). Because penalized estimates are biased, they are not suitable as final inferential quantities.

Accordingly, the study used Group LASSO as a structured feature-selection step before fitting the final survey-weighted logistic regressions. After splitting the data into training and test sets, only the training set was oversampled to address class imbalance. Continuous predictors were standardized and categorical variables were one-hot encoded without dropping reference levels, ensuring that all indicators belonging to the same categorical variable were penalized together. Group LASSO models were then fit across a grid of penalty values (10^{-6} to 10^{-2}) and evaluated on the untouched test set using survey-weighted AUC. For each penalty, group-level L2 norms were calculated and normalized by group size so that multi-level categorical variables could be compared fairly with binary or continuous predictors. These normalized norms, along with trace plots showing each group’s stability across penalties, provided a clear ranking of predictor importance. The penalty that maximized weighted AUC was selected as the optimal model, and the most influential groups at that penalty were retained for the final GLMs. This procedure yielded a consistent, data-driven method for selecting predictors while minimizing the influence of arbitrary coding or group dimensionality.

Second, the reduced predictor set identified by Group LASSO was refit using a survey-weighted logistic regression incorporating the NSDUH analysis weights and Taylor-series

variance estimation. In contrast to the penalized step, the GLM refit used predictors in their conventional form (0/1 dummy indicators with a dropped reference category, and continuous predictors in their original units) to ensure population-representative estimates and interpretable odds ratios. This two-step procedure is well supported as a method for achieving parsimonious models while preserving valid inference under complex survey designs (McConville et al., 2017; Jasiak & Tuvaandorj, 2023).

To support the inferential model, a Random Forest (RF) classifier was used to explore potential nonlinear relationships and interactions that a linear logistic model might miss (Fife & D'Onofrio, 2023; Hastie et al., 2009). Because the study was oriented toward prediction rather than causal inference, the Random Forest served as a complementary analytic tool to identify which combinations of predictors most strongly distinguish treatment access (Fife & D'Onofrio, 2023; Kern et al., 2019; Ryo & Rillig, 2017). The classifier predicted both primary outcomes (past-year receipt of substance use treatment, SUTRTPY, and mental health treatment, MHTRTPY). However, while Random Forests generally excel at prediction and flexibility, they typically do not accommodate the full complexity of multistage survey design in a way that yields valid inferential statistics (Fife & D'Onofrio, 2023; Kern et al., 2019). This limitation arose because RF algorithms inherently lack statistical distribution theory and are non-parametric, which makes it difficult to make sophisticated statistical inferences, such as deriving p-values or confidence intervals from a population distribution (Fife & D'Onofrio, 2023; Ryo & Rillig, 2017). The original RF model was not considered a tool for traditional statistical inference, hypothesis testing, or computing p-values (Ryo & Rillig, 2017). Thus, it was not used to generate reportable odds ratios, standard errors, or p-values. Instead, its role is to benchmark predictive performance (e.g., via Area Under the Curve (AUC), accuracy), and to assess whether

the logistic regression model's performance is adequate or whether important nonlinear patterns remain unmodeled (Kern et al., 2019; Ryo & Rillig, 2017). If the Random Forest substantially outperforms logistic regression, that may motivate further investigation in future work.

Results

Descriptive Statistics

Descriptive characteristics of the analytic sample are summarized in Appendix B, Table B1-B3. Overall, approximately only 5% of respondents reported receiving substance use treatment in the past year, and 26.26% reported receiving mental health treatment. The weighted age distribution was balanced across groups, with the largest proportions falling in the 18-25 (30.10%) and 35-49 (27.37%) age ranges. Females constituted 55.3% of the sample, and non-Hispanic White respondents represented a majority (57.73%), followed by Hispanic (19.50%) and non-Hispanic Black (11.56%) participants. Most respondents reported being never married (46.05%) and insured (89.33%), and roughly 41.41% reported household incomes above \$75,000. With respect to behavioral indicators, 12.85% met criteria for an alcohol-use disorder, 9.78% for a marijuana-use disorder, and 31.33% reported any illicit-drug use in the past year. Attitudinal variables suggested generally high perceived risk: about 65.57% viewed daily cigarette smoking as a “great risk,” whereas perceptions of weekly marijuana use were more evenly distributed. In terms of mental health, 11.49% reported having a major depressive episode in the previous year. These descriptive patterns broadly mirror prior NSDUH findings and indicate a sample that is demographically diverse but largely representative of the civilian, non-institutionalized U.S. population aged 18 and older. Full distributions for all categorical

variables are provided in Appendix B, Table B1-B3, as well as distributions for continuous variables are provided in Appendix B, Figures B1-B3.

Mental Health Treatment (MHTRTPY)

The predictors of past-year mental health treatment showed a clear and fairly consistent pattern across the group LASSO, weighted GLM, and Random Forest models. The group LASSO model (Figures 1-2) achieved a weighted AUC of about 0.74 with accuracy around 0.73, indicating moderate discrimination. At the group level, past-year major depressive episodes (IRAMDEYR) had by far the largest size-normalized L2 norm, followed by race/ethnicity, sex, insurance coverage, illicit drug use, education, and age. Perceived marijuana risk also contributed, but more modestly. In terms of Andersen's Behavioral Model, this pattern suggests that indicators of need (depression and substance involvement) combine with predisposing factors (age, sex, race) and enabling resources (insurance) to shape who receives treatment, with depression standing out as the strongest signal.

The survey-weighted GLM (Figure 3) helps unpack these associations in terms of odds ratios. Adults with a past-year major depressive episode had about six times the odds of receiving mental health treatment compared to those without an episode ($OR \approx 6.17$, 95% CI [5.71, 6.66]). Need-related substance indicators were also important: people who used any illicit drug in the past year had roughly 1.65 times the odds of treatment ($OR \approx 1.65$), and those meeting criteria for marijuana or alcohol use disorders had about 19% and 18% higher odds, respectively. Among sociodemographic factors, women had nearly 1.9 times the odds of treatment compared to men, and having any health insurance increased the odds of treatment by about 79%, reinforcing the role of coverage as a key enabling resource.

The GLM also highlighted substantial disparities across racial and ethnic groups. Compared to non-Hispanic White adults, Black, Hispanic, Asian, and multiracial adults all showed significantly lower odds of receiving treatment, with odds ratios typically between about 0.44 and 0.59, indicating 40-55% lower likelihood of treatment even after adjusting for need and insurance. Age patterns were more nuanced: adults in their late 20s to mid-40s had slightly higher odds of treatment than 18-25-year-olds, whereas adults 65 and older had substantially lower odds ($OR \approx 0.69$). Education showed a mixed pattern, with only those holding advanced degrees clearly more likely to receive treatment ($OR \approx 1.63$), while most other levels were not significantly different from the reference group. Employment and marital status contributed smaller but still meaningful effects: part-time workers and adults not in the labor force had higher odds of treatment than full-time workers, and divorced or separated adults were more likely to receive services than those who were married. Perceived marijuana risk showed a graded pattern where moderate risk perceptions were associated with slightly higher odds of treatment, whereas viewing marijuana as very risky was linked to somewhat lower odds, relative to the lowest-risk category.

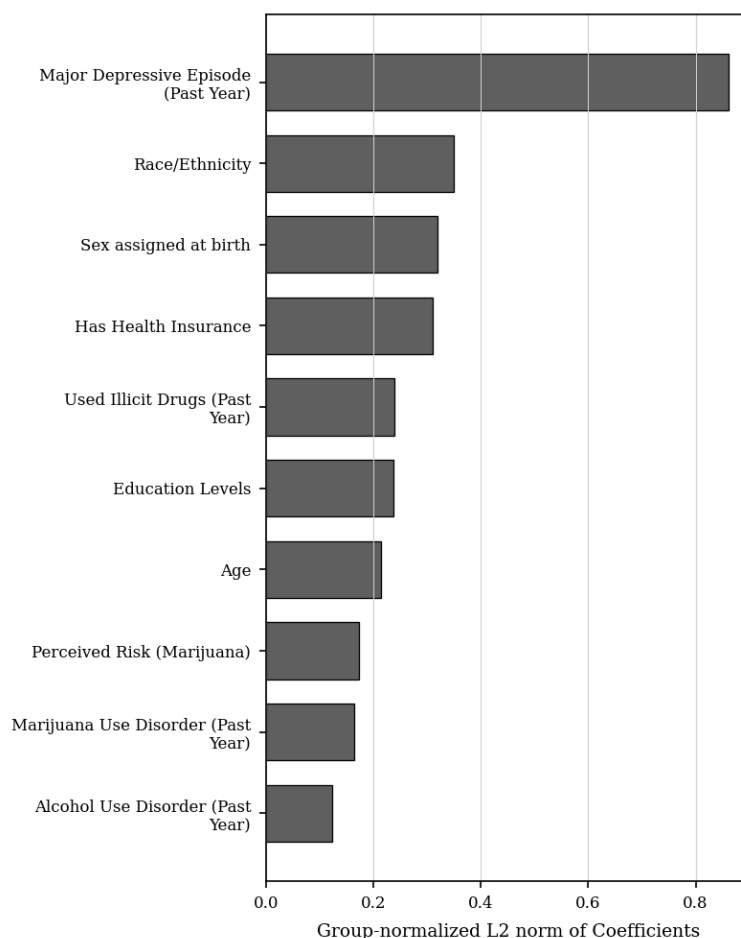
The Random Forest model (Figure 4) emphasized a slightly different set of variables but still pointed to psychological need as central. In the nonlinear model, psychological distress (K6 score) emerged as the most important single predictor, followed by past-year major depressive episodes, alcohol and marijuana use frequency, sex, and illicit drug use. Demographic and socioeconomic variables such as income, education, and employment still contributed, but they carried smaller marginal importance compared to the mental health and substance-related indicators. This pattern suggests that when the model is allowed to capture complex interactions

and nonlinearities, day-to-day distress and recent behavioral health patterns become especially influential.

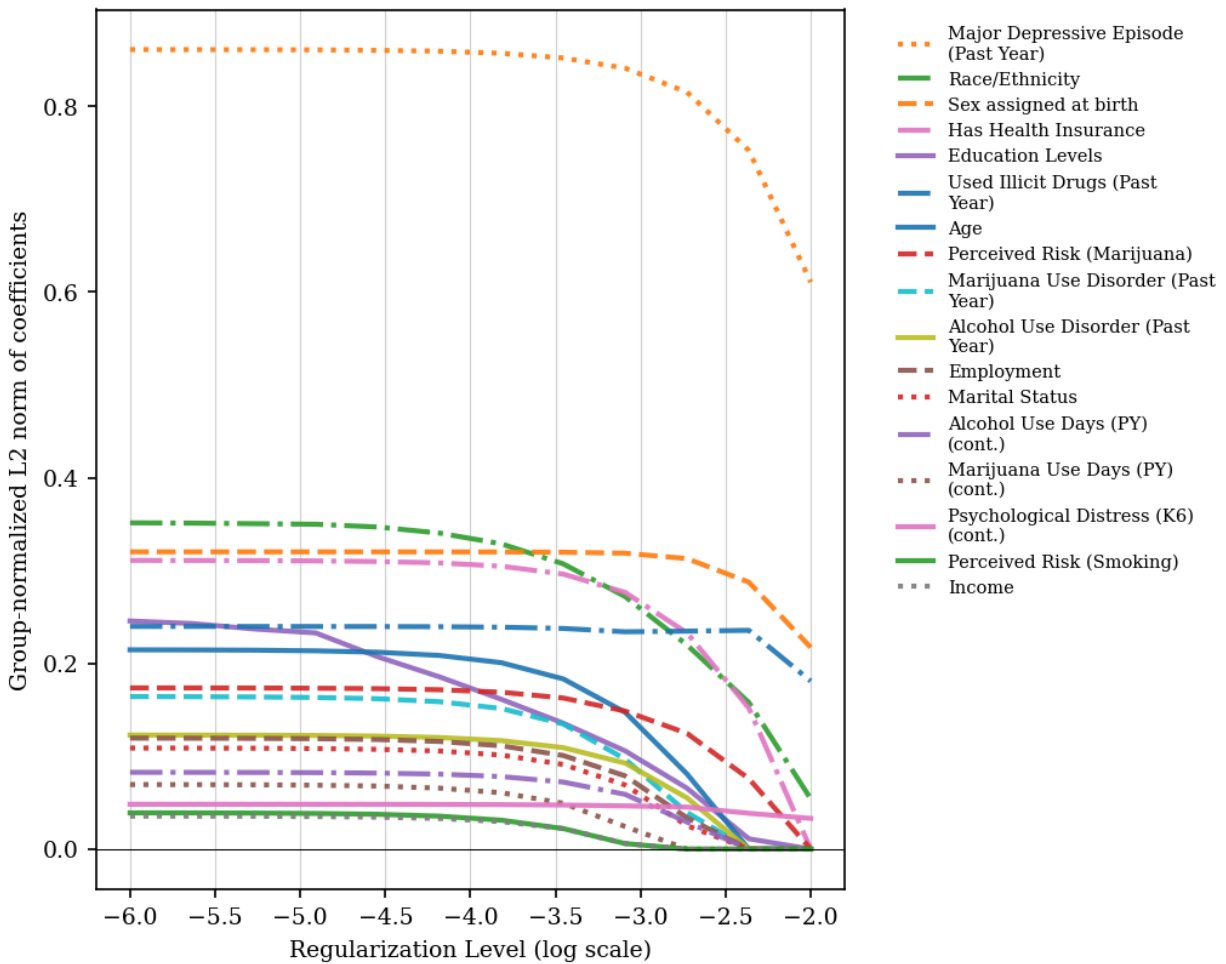
Together, the three modeling approaches converge on a similar story. Past-year depression and other need-related factors (distress and substance involvement) are the strongest and most consistent predictors of mental health treatment, while insurance, sex, and race/ethnicity shape who is actually able to convert that need into care. Linear models (group LASSO and GLM) especially highlight the large gap between adults with and without a recent depressive episode and the persistent racial and ethnic disparities, whereas the Random Forest model gives more weight to continuous distress and use-frequency measures. Overall, mental health treatment utilization among adults appears to be driven by a blend of need, enabling resources, and predisposing social characteristics, with need-based indicators (particularly depression) playing the central role.

Figure 1

Past-Year Major Depressive Episode , Race/Ethnicity. and Sex Were the Most Influential Predictors of Mental Health Treatment



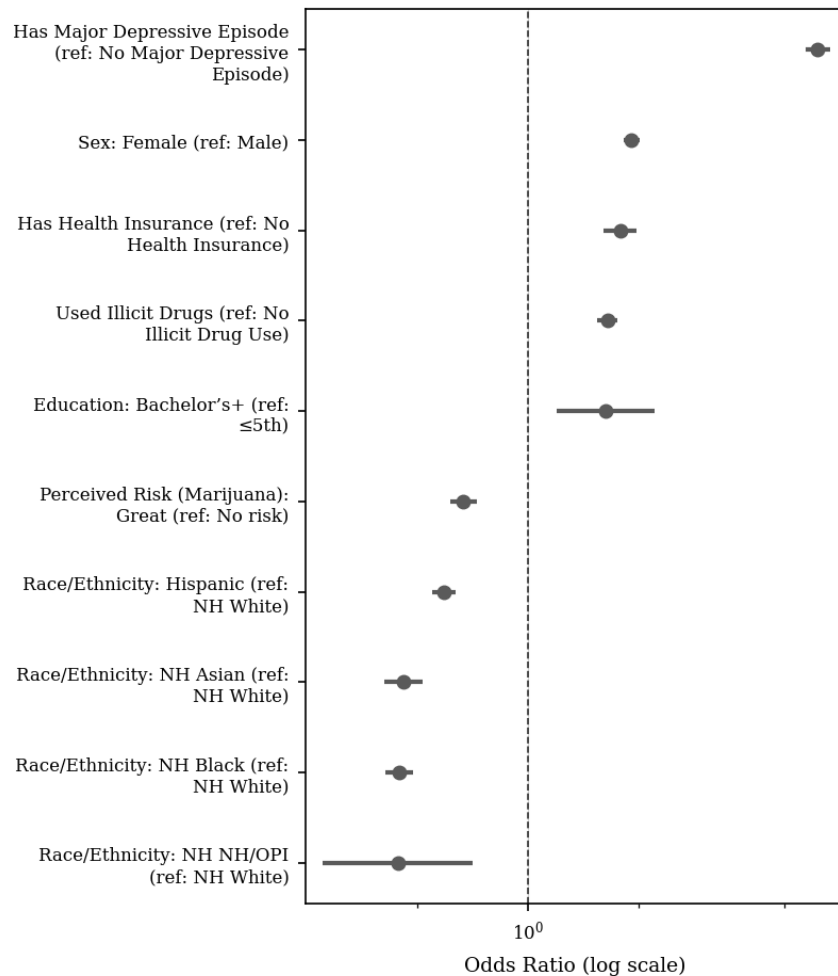
Notes. Predictor groups were selected using a group LASSO model with past-year mental health treatment as the outcome. The plot displays size-normalized L2 norms for each predictor group across the regularization path. Normalizing by group size allows groups with different numbers of dummy variables (e.g., race vs. sex) to be compared on the same scale, highlighting their relative contribution to the model rather than the raw number of coefficients. Full group-level norms and detailed coefficient summaries are reported in Appendix C (Table C1).

Figure 2*Group LASSO Regularization Paths for Mental Health Treatment Predictors*

Notes. Each line shows the group-normalized L2 norm of coefficients across regularization levels in the group LASSO model for past-year mental health treatment. Higher curves indicate predictor groups that remain influential under stronger penalties. Age, race/ethnicity, and education show the strongest and most stable contributions.

Figure 3

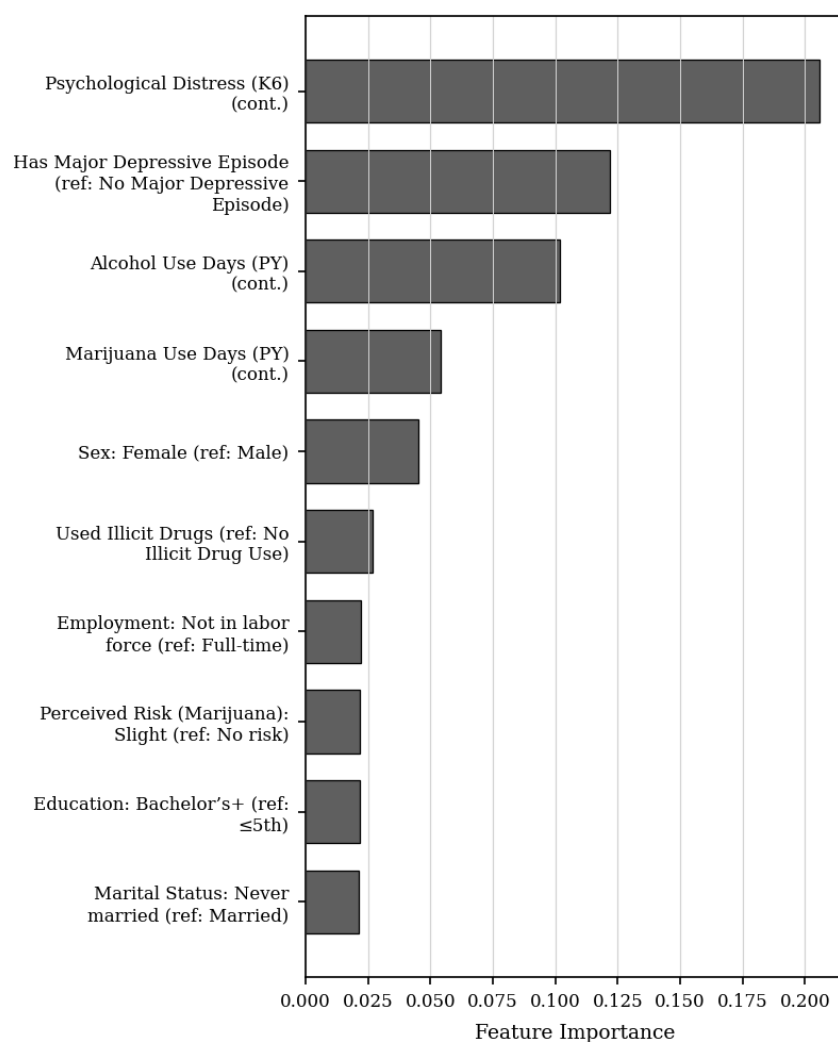
Having Major Depressive Episode, Females, and Having Health Insurance Were Associated With Higher Odds of Receiving Mental Health Treatment



Notes. ref = reference level, NH = Non-Hispanic, NH/OPI = Native Hawaiian/Other Pacific Islander. Error bars represent 95% confidence intervals. Predictor groups were selected using a group LASSO model and refit in a survey-weighted logistic regression model with past-year mental health treatment as the outcome. The plot displays adjusted odds ratios for the final GLM. Full weighted estimates appear in Appendix Table D1-D3.

Figure 4

Psychological Distress, Having Major Depressive Episode, and Alcohol Use Days Were the Most Important Predictors in the Random Forest Model of Mental Health Treatment



Notes. ref = Reference level being dropped, cont. = Continuous variable, PY = Past Year, Income is displayed in US Dollars. Bars display mean decrease in Gini impurity from the Random Forest model; larger values reflect greater predictive importance. Variables were selected from a Random Forest model having mental health treatment in the past year as the outcome variable. Full coefficient values for each level can be found in appendix E, table E1-E4.

Substance Use Treatment (SUTRTPY)

The predictors of past-year substance use treatment also showed strong convergence across the group LASSO, survey-weighted GLM, and Random Forest models. The group LASSO model (Figures 5-6) achieved a weighted AUC of about 0.78 with accuracy around 0.78, suggesting moderately strong discrimination despite the rarity of treatment in the sample. Using size-normalized L2 norms, the most influential groups were alcohol use disorder, past-year major depressive episode, age, illicit drug use, and education, followed by marijuana use disorder, income, marital status, and insurance. Race/ethnicity and sex contributed smaller but still noticeable signal. This pattern again lines up with Andersen's Behavioral Model: need-related behavioral and clinical indicators carry most of the predictive weight, while predisposing (age, race) and enabling factors (income, insurance, employment) shape who actually receives care.

The survey-weighted GLM (Figure 7) clarifies these relationships in terms of odds ratios. Respondents with a past-year alcohol use disorder had more than twice the odds of receiving substance use treatment compared to those without ($OR \approx 2.32$, 95% CI [2.07, 2.60]). Similarly, those who used any illicit drug in the past year had about 2.39 times the odds of treatment ($OR \approx 2.39$), and people meeting criteria for marijuana use disorder had roughly 55% higher odds ($OR \approx 1.55$). Past-year major depressive episodes also played a major role: adults with a recent MDE had almost three times the odds of receiving substance use treatment ($OR \approx 2.93$), highlighting the importance of psychiatric comorbidity in treatment entry.

Age patterns were pronounced. Using 18-25-year-olds as the reference group, adults in their late 20s to mid-30s had about 1.83 times the odds of treatment, those 35-49 had roughly 2.68 times the odds, and adults 50-64 had about 1.93 times the odds. Adults 65 and older did not differ significantly from the youngest adults. Education was also strongly associated with

treatment, but in a non-linear way. Compared to adults with ≤ 5 th-grade education, those with 7th–8th grade, 9th–11th grade, and various high school or some-college levels often showed two- to five-fold increases in the odds of treatment (e.g., $OR \approx 5.15$ for 7th–8th grade; $OR \approx 4.47$ for a GED). At the same time, income moved in the opposite direction: compared to the lowest income group, adults in the highest income bracket (\$75,000+) had about 46% lower odds of receiving treatment ($OR \approx 0.54$), with a clear monotonic gradient across the middle income categories as well ($ORs \approx 0.69$ – 0.63). These patterns suggest that people with mid-range educational attainment but fewer financial resources may be more likely to encounter mandated, safety-net, or publicly funded treatment settings.

Other demographic and structural variables also mattered. Several racial and ethnic minority groups had significantly lower odds of treatment than non-Hispanic White adults, even after adjusting for clinical need. For example, some groups showed odds ratios in the range of 0.62–0.84, indicating roughly 15-40% lower likelihood of receiving care. Unlike the mental health treatment model, women were slightly less likely than men to receive substance use treatment ($OR \approx 0.88$), pointing to a gender gap that runs in the opposite direction. Insurance still appeared as an important enabling resource: having any health insurance was associated with about 1.71 times the odds of treatment. Employment and marital status showed additional differences, with part-time workers, unemployed adults, and those not in the labor force having higher odds of treatment than full-time workers ($ORs \approx 1.33$ - 1.46), and widowed and divorced/separated adults showing substantially higher odds than married adults ($ORs \approx 1.51$ - 1.99).

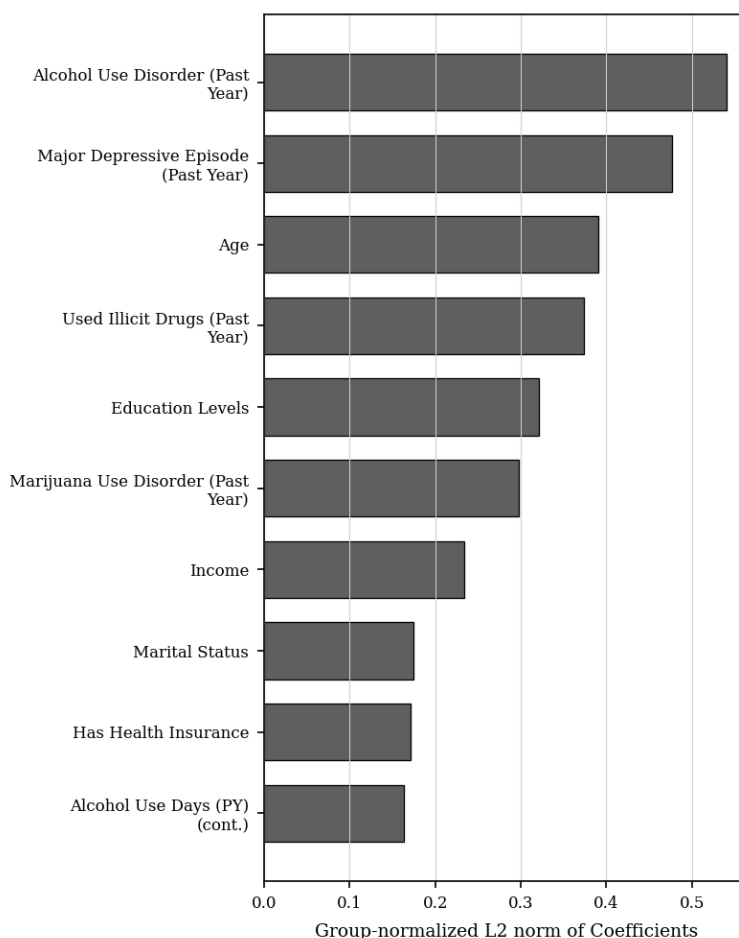
The Random Forest model (Figure 8) further emphasized the centrality of behavioral indicators and severity. In this nonlinear model, alcohol use frequency (IRALCFY) and

psychological distress (K6 score) emerged as the two most important features, followed by marijuana use frequency (IRMJFY) and clinical indicators such as alcohol use disorder, illicit drug use, and marijuana use disorder. Demographic and socioeconomic variables (such as sex, income, employment, and marital status) still contributed, but they generally had smaller importance scores compared to the heavy emphasis on substance involvement and distress. Perceived risk variables (e.g., marijuana and cigarette risk) showed only modest importance, suggesting that attitudinal factors are less central than concrete behavioral patterns when it comes to predicting who ends up in treatment.

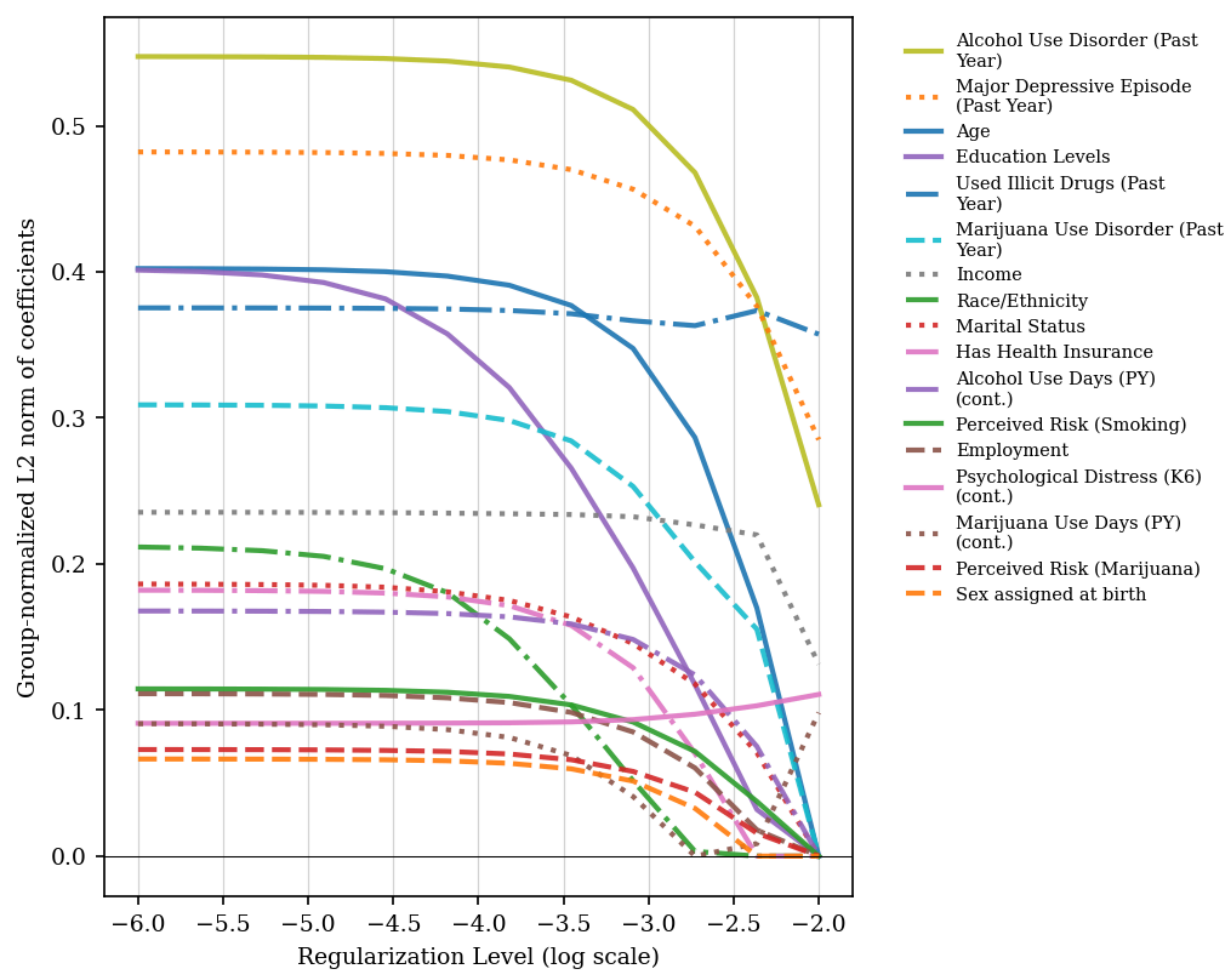
Overall, the three modeling approaches tell a consistent story for substance use treatment. Utilization is primarily driven by behavioral and clinical markers of need (particularly alcohol involvement, illicit and marijuana use, and co-occurring depression) while sociodemographic characteristics and enabling resources shape the chances of converting that need into actual treatment. Compared to the mental health treatment model, the substance use model places even more weight on heavy use and comorbidity and reveals a different gender pattern, with men more likely to receive SUD services. Together, these findings suggest that in the general adult population, entering substance use treatment is closely tied to observable severity and multi-faceted substance involvement, layered on top of persistent disparities by race, sex, and socioeconomic position.

Figure 5

Past Year Alcohol Use Disorder, Past Year Major Depressive Episode, Age Were the Most Influential Predictors of Substance Use Treatment



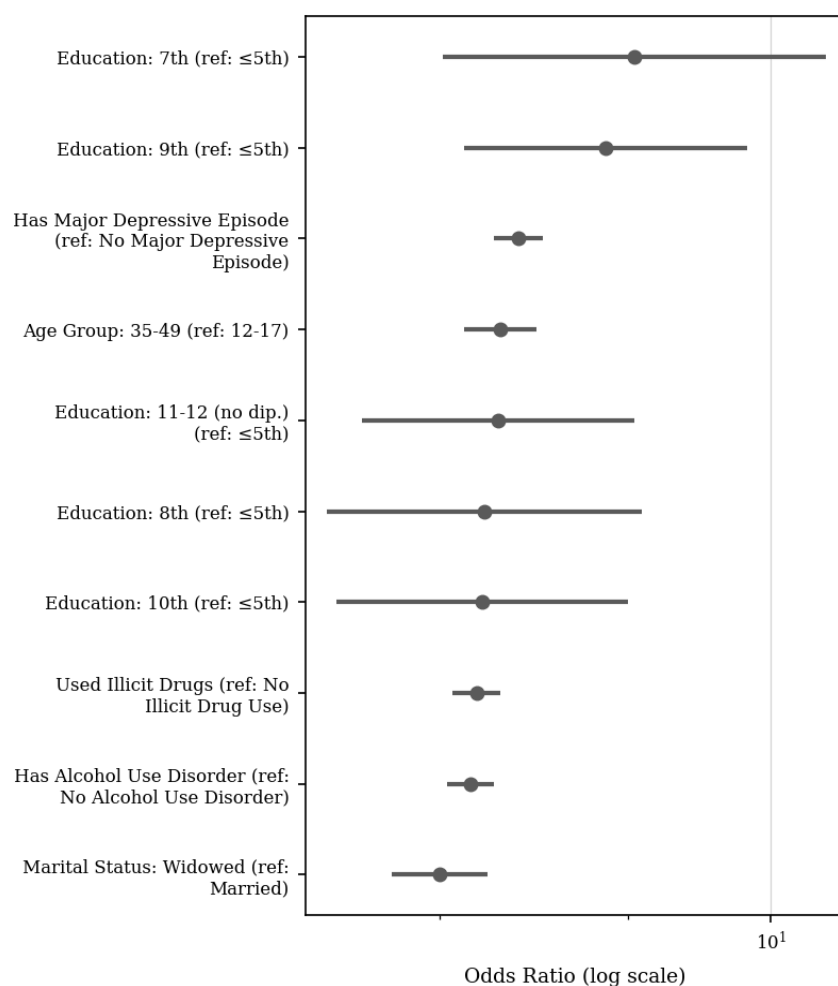
Notes. Predictor groups were selected using a group LASSO model with past-year substance use treatment as the outcome. The plot displays size-normalized L2 norms for each predictor group across the regularization path. Normalizing by group size allows groups with different numbers of dummy variables (e.g., race vs. sex) to be compared on the same scale, highlighting their *relative* contribution to the model rather than the raw number of coefficients. Full group-level norms and detailed coefficient summaries are reported in Appendix C (Table C2).

Figure 6*Group LASSO Regularization Paths for Substance Use Treatment Predictors*

Notes. Each line shows the group L2 norm of coefficients across regularization levels in the group LASSO model for past-year mental health treatment. Higher curves indicate predictor groups that remain influential under stronger penalties. Education, alcohol use disorder, and age show the strongest and most stable contributions.

Figure 7

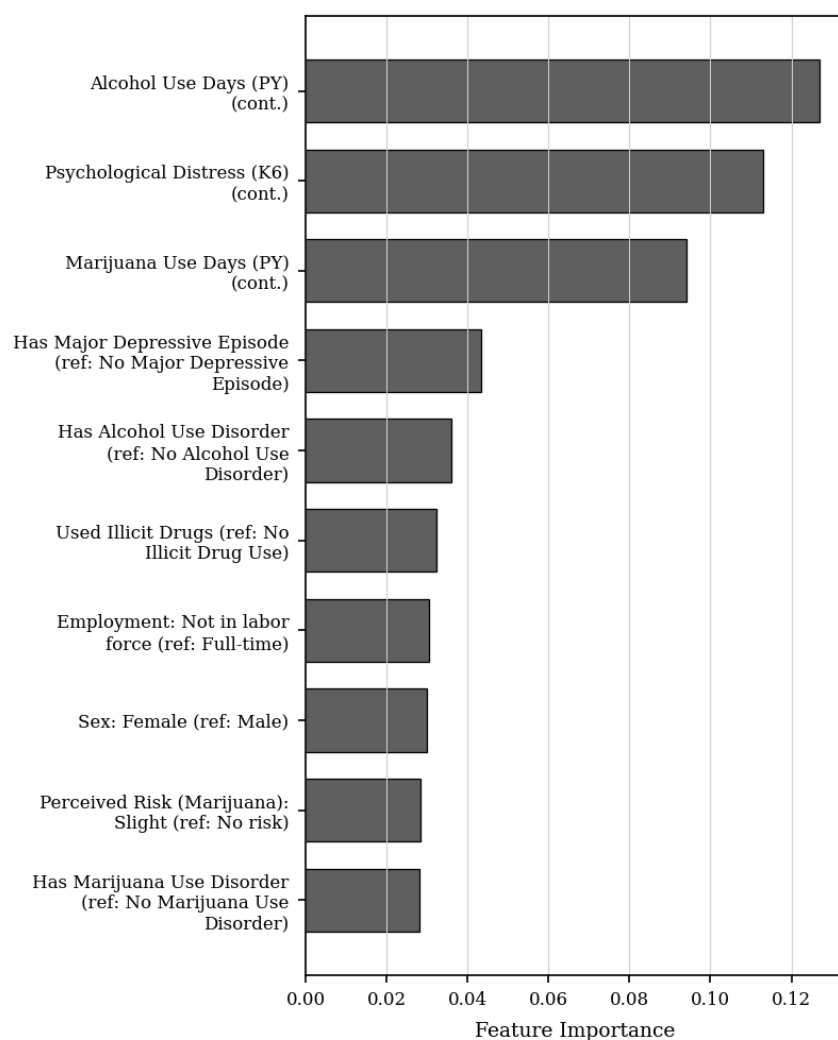
Mid-Range Education Levels, Having Major Depressive Episode, and Adults aged 35-49 Were Associated With Higher Odds of Substance Use Treatment



Notes. ref = Reference level being dropped, no dip. = no diploma, Income is displayed in US Dollars. Error bars reflect 95% confidence intervals. Predictor groups were selected using a group LASSO model and then refit in a survey-weighted logistic regression model with past-year substance use treatment as the outcome. The figure displays odds ratios for the final GLM, with full weighted estimates reported in Appendix Table D4-D6.

Figure 8

Alcohol Use Days, Psychological Distress, and Marijuana Use Days Were the Most Important Predictors of Substance Use Treatment in the Random Forest Model



Notes. ref = Reference level being dropped, cont. = Continuous variable, PY = Past Year. Bars display mean decrease in Gini impurity from the Random Forest model; larger values reflect greater predictive importance. Variables were selected from a Random Forest model having substance use treatment in the past year as the outcome variable. Full coefficient values for each level can be found in appendix E, table E5-E8.

Discussion

The present study sought to examine the factors that predict behavioral-health treatment utilization in the United States, focusing on both mental health and substance use treatment. Drawing on Andersen's Behavioral Model of Health Services Use and Ajzen's Theory of Planned Behavior, the study tested three primary hypotheses: that (1) structural enablers such as insurance, income, and education, along with clinical indicators of need, would be the strongest predictors of treatment utilization; (2) demographic and attitudinal variables, including gender, race/ethnicity, and perceived risk, would moderate these relationships, contributing to persistent underutilization among men and racial/ethnic minorities; (3) age would emerge as an independent and underexamined factor, with older adults exhibiting lower treatment utilization after accounting for other predictors; and (4) predictor overlap would be substantial across both mental health and substance use treatment, with many of the same structural, demographic, and clinical factors shaping engagement in both service systems.. The findings broadly supported these expectations, with some important nuances across treatment domains and modeling approaches. Clinical indicators of need and core structural enablers were consistently among the strongest predictors, especially for substance use treatment, confirming that both severity and opportunity drive access to care. Demographic variables also shaped utilization in predictable ways: women had substantially higher odds of receiving mental health treatment, whereas men were more likely to receive substance use treatment, and racial and ethnic minorities remained underrepresented in both domains. Age emerged as a meaningful factor, with older adults less likely to receive mental health treatment and treatment for substance use concentrating in midlife. Attitudinal variables, such as perceived risk of marijuana use, played a more modest role and were overshadowed by need-based and structural predictors, particularly in the substance use

models. And through the Group LASSO, GLM and Random Forest analyses, overlapping predictors (such as past-year major depressive episodes) were showing significant results on both models of the two types of treatments.

Theoretical Implications

Demographic: Predisposing factors

Across models, the most influential demographic predictors reflected predisposing pathways that shape how individuals interpret symptoms and assess the acceptability of help-seeking. Female gender emerged as one of the most consistent positive predictors of mental health treatment: in the weighted GLM, women had nearly twice the odds of receiving care compared to men, and this pattern remained stable in the group LASSO and Random Forest models. From a theoretical perspective, this aligns with both Andersen's and Ajzen's frameworks. Under Andersen's model, gender functions as a predisposing characteristic that shapes perceptions of need and orientation toward professional care. Under TPB, gender differences map onto attitudes, subjective norms, and perceived behavioral control: women tend to report greater emotional openness, lower stigma, and greater willingness to acknowledge distress, all of which strengthen the intention pathway to treatment.

By contrast, lower utilization among older adults and several racial and ethnic minority groups reflects a different set of predisposing influences. Even after adjusting for insurance, education, employment, and clinical need, Black, Hispanic, Asian, and multiracial adults showed significantly lower odds of receiving mental health treatment. These disparities are consistent with Andersen's proposition that social position shapes both perceived and evaluated need, and they also map onto TPB constructs: communities facing historical discrimination or cultural mistrust of formal systems often experience stronger negative subjective norms and weaker

perceived behavioral control. In this sense, lower treatment rates among minority groups likely represent a combination of reduced enabling opportunities (e.g., provider availability, culturally responsive care) and attitudinal barriers rooted in stigma, expectations of self-reliance, and concerns about credibility or misdiagnosis. Together, these patterns show that predisposing factors in mental health care operate primarily through beliefs, norms, and sociocultural interpretations of distress, not simply through demographic differences alone.

For substance use treatment, predisposing factors appeared to operate through a different mechanism: one more closely tied to life context, system contact, and external pressure than to internal motivation. Treatment was heavily concentrated among middle-aged adults, especially those in their mid-30s to late 40s, who had substantially higher odds of receiving substance use treatment than young adults. In Andersen's terms, age functions as a predisposing characteristic that interacts with need severity and enabling context: middle adulthood is the period when substance use often produces employment, health, or family consequences that increase both the visibility of need and the likelihood of referral or intervention. Within TPB, this pattern suggests that behavioral intention may be less central for SUD treatment because many individuals enter care through court mandates, workplace referrals, or family pressure: normative forces that bypass or override personal attitudes.

Racial and ethnic disparities again emerged, with several non-White groups showing lower odds of treatment even after controlling for substance use disorder severity and depression. This reinforces the role of predisposing structural disadvantage and culturally shaped norms surrounding addiction, shame, and mistrust. Unlike the mental health models, gender reversed direction: women were slightly less likely than men to receive substance use treatment, highlighting how TPB norms around addiction, motherhood, and moral judgment may strongly

inhibit perceived behavioral control and willingness to disclose problems. In Andersen's framework, these results point to persistent inequities in enabling resources such as referral pathways, culturally tailored treatment options, and provider engagement that disproportionately affect minority women.

Socioeconomic: Enabling factors

For mental health treatment, enabling factors functioned as fairly straightforward gateways. Having any form of health insurance was strongly associated with treatment use in the GLM, and insurance also had a large group-level contribution in the LASSO models. These patterns underscore the central role of coverage in accessing outpatient and therapeutic services. Employment and education showed smaller and more mixed associations: part-time workers and those not in the labor force had somewhat higher odds of treatment than full-time workers, and individuals with advanced degrees had elevated odds compared to those with the lowest education, but many intermediate education levels did not differ significantly. Income was less central in the mental health models than in the substance use models and did not appear in the final GLM. Overall, these results still support Andersen's assertion that enabling resources facilitate the movement from perceived need to actual service use, especially through insurance coverage and basic socioeconomic stability.

For substance use treatment, enabling factors mattered but were more complex and sometimes counterintuitive. Insurance again emerged as a positive predictor, with insured adults significantly more likely to receive treatment than uninsured adults. However, patterns for education, employment, and income were more nuanced. Several mid-range education categories (for example, 7th–10th grade or GED-equivalent levels) were associated with dramatically higher odds of treatment compared to the lowest education group, while the effects for higher

education were less consistent. Employment status also mattered: people employed part time, unemployed, or not in the labor force had higher odds of treatment than full-time workers. In contrast, income showed a clear inverse gradient: adults in higher income categories, especially those earning \$75,000 or more, were substantially less likely to receive substance use treatment than those in the lowest income group. These patterns are consistent with treatment pathways that are at least partly driven by mandated or crisis-oriented systems (e.g., criminal justice referrals, child welfare involvement, or safety-net programs) rather than purely voluntary help-seeking. In other words, socioeconomic factors may increase voluntary access in mental health care but also structure institutional and coercive pathways into substance use treatment.

Behavioral and Clinical: Need factors

For mental health treatment, the clearest and most robust indicator of need was past-year major depressive episode (IRAMDEYR). In the survey-weighted GLM, experiencing a past-year episode increased the odds of treatment by roughly a factor of six, making it the single strongest predictor in the linear models. The group LASSO also identified IRAMDEYR as the dominant group-level contributor. Comorbid substance involvement (especially any illicit drug use and the presence of alcohol or marijuana use disorders) also predicted higher mental health treatment utilization, though with more modest effects. In the Random Forest model, continuous psychological distress (K6) emerged as the top predictor, followed by depression, alcohol and marijuana use frequency, and substance use disorders. This suggests that when the model can flexibly capture nonlinear relationships, day-to-day distress and recent use patterns become especially influential. At the same time, the mental health models showed that need alone is not sufficient: the strongest effects for depression and substance use occurred in the context of

insurance and other enabling resources, reinforcing Andersen's argument that need must intersect with structural pathways to result in actual treatment.

For substance use treatment, need indicators overwhelmingly dominated the prediction landscape. In the GLM and group LASSO models, alcohol use disorder, marijuana use disorder, any past-year illicit drug use, and major depressive episode all showed large and significant effects, with odds ratios often between about 1.5 and 3.0 relative to those without these conditions. The Random Forest model further emphasized behavioral severity: alcohol and marijuana use frequency, distress, and substance use disorders ranked among the most influential features, ahead of most demographic and socioeconomic variables. These patterns suggest that substance use treatment is closely tied to observable and often severe behavioral involvement, which may be detected not only by individuals themselves but also by families, workplaces, and systems such as the criminal justice system. In this domain, "evaluated need" is frequently defined by external institutions in addition to (or instead of) self-perceived need, refining Andersen's model to account for externally imposed thresholds for care.

Attitudinal and Risk-Perception Factors

For mental health treatment, perceived risk of marijuana use showed small but meaningful associations. In the GLM, some levels of marijuana risk perception were linked to slightly higher or lower odds of treatment relative to the lowest-risk group, and group LASSO and Random Forest models also assigned non-zero importance to these variables. These patterns suggest that broader health-related caution or preventive orientation may support help-seeking to some extent. In Ajzen's terms, such variables likely reflect general attitudes toward health behavior and perceived behavioral control, where seeing risky behaviors as harmful and

treatment as accessible makes it easier to translate distress into action. Still, the effect sizes were modest compared to depression and substance involvement.

For substance use treatment, attitudinal factors were even weaker. Perceived risk of marijuana and cigarettes contributed small signals in the group LASSO and Random Forest models but did not appear as core predictors in the final GLM. Individuals who endorsed higher risk perceptions showed slightly greater likelihood of reporting treatment in some models, but these associations were dwarfed by indicators such as alcohol use disorder, multi-substance involvement, and co-occurring depression. This likely reflects the reality that substance use treatment is often crisis-driven or mandated, leaving less room for personal attitudes and intentions to shape the initial decision to seek care.

Practical Implications

Although this study was primarily theoretical and data-driven, the findings have several practical implications for addressing behavioral-health disparities in the United States. Across both treatment domains, consistent patterns emerged for specific population groups: youth, older adults, racial and ethnic minorities, low-income adults, and individuals with co-occurring conditions. These groups exhibited distinct profiles of need, access, and treatment entry. Organizing implications around these populations helps clarify where targeted interventions may have the greatest impact.

Mental Health and Substance Use as Parallel Systems

One of the main takeaways from this study is that several of the strongest predictors: major depressive episode, psychological distress, alcohol use disorder, marijuana involvement, and illicit drug use showed up in both the mental health and substance use treatment models. This overlap suggests that the two treatment systems are not as separate as they often appear.

Instead, many adults come in with overlapping needs, and the same symptoms that push someone toward mental health care also tend to signal problems in the substance use domain. This fits with a large body of research showing how common co-occurring disorders are. For example, almost half of people with a lifetime substance use disorder also have a lifetime mental health disorder (Harris & Edlund, 2005), and dual-diagnosis cases tend to have more severe symptoms and higher risks of hospitalization or self-harm (Sullivan, 2022). The fact that these clinical indicators predict both treatment types also lines up with Andersen's model, which emphasizes the central role of need, and with TPB, which explains how distress and impairment shape people's motivation to seek help.

Past studies also show that people with co-occurring conditions often move between systems or get stuck in the gaps between them. Some individuals with both mental health and substance use problems are more likely to show up in SUD programs (Chen et al., 2013), while others prefer mental health services even when substance use is a major issue (Sidahmed et al., 2015). At the same time, many SUD providers end up delivering mental health interventions without a coordinated system behind them (Weaver et al., 2003). These mixed patterns mirror our findings: clinical severity pushes people toward some form of treatment, but the specific system they enter often depends on what feels more accessible, acceptable, or familiar.

Overall, these results point to the need for more integrated and person-centered approaches. Cross-screening, such as asking about substance use in mental health settings and screening for depression in SUD programs, could help identify unmet needs earlier. Likewise, shared treatment environments or coordinated teams could prevent people with complex needs from bouncing between disconnected services. From a theoretical perspective, Andersen's model helps explain why overlapping clinical need leads to similar treatment predictors across systems,

and TPB helps explain why attitudes, norms, and perceived control influence which system a person turns to first. In short, treating mental health and substance use as separate worlds doesn't match how people actually experience these problems, and our findings highlight why a more integrated behavioral-health approach is necessary.

Implications for Age Groups

Age-related findings revealed a pronounced gap among older adults, who consistently showed lower utilization of both mental health and substance use treatment despite experiencing substantial distress and behavioral health concerns. Within Andersen's framework, age operates as a predisposing characteristic that shapes beliefs about illness, expectations about care, and the likelihood of interpreting symptoms as warranting professional help. From a TPB perspective, generational norms surrounding emotional restraint, independence, and self-management may suppress attitudes favorable to treatment and weaken perceived behavioral control. Practical barriers: reduced mobility, limited awareness of resources, or difficulty navigating complex systems further constrain the enabling-resource pathway. Given these theoretical patterns, interventions should target both perceptions and structural access. Integrating behavioral-health screening into primary care can help shift the evaluated need for older adults who may under-recognize symptoms. Expanding tele-mental health options may enhance perceived control, while age-tailored communication strategies can reshape attitudes and subjective norms about help-seeking in later life.

Implications for Racial and Ethnic Minority Groups

Racial and ethnic minority groups showed consistent underutilization across both treatment types. Black, Hispanic, and Asian adults all had significantly lower odds of receiving

care compared to White adults, even after accounting for insurance, income, and clinical need. In Andersen's model, these disparities reflect the dual influence of predisposing cultural orientations and enabling structural constraints. However, TPB helps clarify why structural expansion alone is insufficient: cultural mistrust, experiences of discrimination, and concerns about being misunderstood or misdiagnosed shape subjective norms and attitudes toward professional treatment. These factors weaken behavioral intentions even when formal access is available. Therefore, closing racial and ethnic treatment gaps requires interventions that address both enabling resources and belief systems. Culturally adapted services, language-concordant providers, and improved cultural humility within the workforce can strengthen perceived behavioral control and reduce mistrust. Partnerships with trusted community institutions, such as churches or cultural organizations, can shift subjective norms in supportive ways. Ultimately, expanding insurance coverage must be paired with culturally responsive engagement strategies if disparities are to be meaningfully reduced.

Implications for Low-Income and Uninsured Individuals

Socioeconomic disadvantage played a substantial role in treatment patterns, especially for mental health services. Low-income and uninsured adults were significantly less likely to receive mental health treatment, underscoring Andersen's claim that enabling resources (insurance, financial stability, and service availability) function as gatekeepers to care. These patterns also intersect with TPB: when individuals anticipate high costs, long wait times, or difficulty navigating coverage, their perceived behavioral control over accessing care diminishes, lowering the intention to seek treatment even when need is high. Policies that expand Medicaid eligibility, reduce out-of-pocket costs, and support community-based or telehealth services would help mitigate these enabling barriers. Yet structural interventions should be paired with efforts to

improve insurance literacy, as many adults remain unaware of available coverage or mental health benefits.

Substance use treatment revealed a more complex socioeconomic pattern. Adults not in the labor force or with mid-range education levels were more likely to receive treatment, suggesting pathways shaped by external forces (such as criminal justice referrals, crisis stabilization units, or mandated programs) consistent with Andersen's recognition that utilization is not always voluntary. Higher-income adults showed lower treatment rates, indicating differing social consequences, private resources, or less visible impairment. These findings emphasize that enabling resources influence treatment differently depending on whether help-seeking is internally motivated (TPB-driven) or externally compelled (structurally driven), highlighting the need for context-specific policy approaches.

Strengths and Limitations

Strengths

This study offered several notable strengths that make it a valuable contribution to the behavioral-health literature. One of its primary advantages was the use of a large, nationally representative dataset. By drawing on data from the 2023 National Survey on Drug Use and Health (NSDUH), the study captured a broad snapshot of the U.S. population, encompassing both adolescents and adults across diverse social, economic, and geographic backgrounds. Because NSDUH employed complex sampling and weighting procedures, the resulting estimates accurately reflect national trends rather than isolated samples.

Another strength was in the study's theoretical integration: this research was guided by two well-established frameworks: Andersen's Behavioral Model and Ajzen's Theory of Planned

Behavior (1985). Grounding the analytic approach in theory provided interpretive clarity and ensured that statistical findings could be meaningfully connected to social and psychological mechanisms. In addition, the study's comparative design: analyzing both mental health and substance use treatment within the same framework, offered an important extension to previous research. Prior national analyses, such as those by Wu et al. (2017) and Han et al. (2020), typically examined these outcomes in isolation, leaving unclear whether the same determinants applied across both domains. By analyzing them side-by-side, this project illustrates both their overlaps and their distinctions.

A further strength of this study was in its methodological design. The combined use of group LASSO, survey-weighted logistic regression, and Random Forest modeling allowed the analysis to capitalize on the strengths of three distinct statistical frameworks. Rather than relying on a single model, the study used each technique for a specific analytic purpose: group LASSO provided data-driven feature selection at the level of conceptual domains; the weighted GLM offered interpretable, population-level estimates with proper accommodation of survey design; and the Random Forest model revealed nonlinear associations and interaction patterns that traditional regression approaches may overlook. This triangulated modeling structure helped minimize the limitations of any one method and increased confidence in the stability and generalizability of the findings. When predictors consistently emerged as influential across all three approaches, the convergence itself strengthened the substantive conclusions.

Another methodological strength was the explicit use of survey weights from NSDUH, which ensured that the estimates reflected nationally representative patterns rather than sample-specific artifacts. By integrating weights directly into the GLM and contextualizing the LASSO and Random Forest findings with weighted outcome patterns, the study preserved the

population validity of its conclusions. In addition, the decision to examine predictor domains rather than only individual variables represents a conceptual strength. The group LASSO framework enabled predictors to be evaluated in clusters aligned with Andersen's model, thereby grounding the analysis in theory rather than treating variables as isolated pieces of information. This domain-based approach also helped differentiate strong, stable predictors (such as sex, insurance, and indicators of behavioral need) from groups that contributed weak or inconsistent signal across models. Furthermore, the study's treatment of age as a substantive predictor, rather than a routine control variable, provided an important contribution to the behavioral-health literature. Across models, age demonstrated a nonlinear, life-course pattern, peaking in young and middle adulthood and declining sharply among older adults. Highlighting these generational differences advances a more dynamic understanding of treatment utilization and underscores the importance of developmental context in health-services research.

Limitations

Despite its contributions, the study has several methodological and technical limitations that should temper interpretation. Although the combined use of group LASSO, survey-weighted logistic regression, and Random Forest modeling strengthened the robustness of the findings, each approach carries intrinsic constraints. Group LASSO, for example, is known to introduce bias through coefficient shrinkage, particularly when predictors are correlated or when groups vary substantially in size. Even with size-normalized L2 norms, some categorical variables contained many more levels than others, meaning that groups with large numbers of dummy indicators may still behave differently in the penalty structure than smaller groups. This uneven granularity introduces subtle interpretive challenges and may affect how strongly different variables appear in the regularization path. At the same time, refitting the selected predictors in a

survey-weighted GLM reintroduces the possibility of multicollinearity, because the GLM does not inherit LASSO's regularization. This trade-off between bias reduction, feature stability, and multicollinearity is inherent to post-selection inference, and although model diagnostics confirmed sufficient rank and stability, some standard errors may remain inflated.

The study's cross-sectional design further imposes a fundamental limitation on causal inference. Because NSDUH measures all variables at a single time point, the models can identify associations but cannot determine temporal ordering. For example, the analyses showed strong links between treatment utilization and factors such as income, insurance coverage, and psychological distress, but it remains unclear whether these variables shaped entry into treatment or whether receiving treatment itself influenced changes in employment status, distress levels, or insurance enrollment. Similar ambiguities apply to attitudinal variables and substance use behaviors. As noted in prior national survey research (Walker et al., 2015; Ali et al., 2015), temporal ambiguity limits the ability to draw conclusions about mechanisms or directional pathways.

Another constraint involves the rarity of substance use treatment in the dataset. Only about 1–2% of adults reported receiving treatment in the past year, creating substantial class imbalance. Although the study used oversampling to stabilize machine-learning predictions and applied survey weights to preserve population representativeness, rare-event bias may still reduce sensitivity, particularly in smaller demographic subgroups. Even Random Forest models (which typically handle imbalance more effectively) can overweight high-frequency predictors or miss associations that are real but subtle. The rarity of treatment also means that model estimates for some education levels, racial/ethnic subgroups, or employment categories may have wide confidence intervals and should be interpreted cautiously.

As mentioned in the Methods section, the decision to restrict the analytic sample to adults aged 18 and older further limits the generalizability of the findings. This restriction was necessary because several key predictors were not measured consistently among adolescents. However, removing youth from the study means that the findings describe only adult pathways into treatment and cannot speak to early intervention, adolescent help-seeking, or transitions from youth to adult behavioral-health systems. Patterns of utilization (and the barriers shaping them) may differ substantially for younger populations, and future research should incorporate developmental context into models of behavioral-health service use.

The study also relies entirely on self-reported data, which are subject to recall bias, social desirability effects, and underreporting especially in areas involving stigma, such as illicit drug use or mental health treatment. Measurement error likely attenuates relationships between predictors and treatment outcomes and may obscure associations in subgroups with lower disclosure rates. Additionally, although the analytic approaches showed broad agreement, Random Forest variable importance reflects predictive contribution rather than causal influence, and GLM estimates should be interpreted as associations rather than estimates of causal effects.

Finally, the structure of NSDUH itself presents limitations. The survey measures whether treatment occurred but not its duration, intensity, modality, appropriateness, or quality. As a result, the study cannot distinguish between brief contact, mandated interventions, crisis-driven episodes, or sustained therapeutic engagement. This limitation, while typical of federal surveillance systems, means that the findings speak primarily to initiation of treatment rather than the full continuum of behavioral-health care.

Threats to External Validity

Although the NSDUH is designed to be nationally representative of the civilian, non-institutionalized U.S. population, several factors limit the broader generalizability of these findings. Because the survey excludes institutionalized groups including incarcerated individuals, active-duty military members, and residents of long-term psychiatric or residential treatment facilities, the results may understate disparities at the most acute ends of behavioral-health need. These excluded populations exhibit elevated rates of both mental illness and substance use disorders, and their pathways into treatment often differ substantially from those of community-dwelling adults. Another important limitation involves the restriction of this study to adults aged 18 and older, due to the absence or inapplicability of several predictors for adolescents (e.g., K6 distress, employment status). While analytically necessary, this decision means that the findings cannot be generalized to youth treatment-seeking, early intervention patterns, or the developmental processes that precede adult service use. Behavioral-health barriers in adolescence may differ qualitatively from those observed in adults, and the present results speak only to adult pathways into care.

Temporal and cultural boundaries also constrain external validity. The data reflect conditions in 2023, a period marked by ongoing shifts in telehealth access, economic stability, and public attitudes toward behavioral health. Changes in policy environments for example, the expansion of tele-mental health after the COVID-19 pandemic may alter treatment accessibility in ways not captured here. Moreover, generational differences may shape help-seeking independently of chronological age; older and younger adults may differ because of distinct cultural experiences with stigma, technology, or healthcare messaging.

Finally, NSDUH's suppression of geographic identifiers in the public-use file limits the ability to model regional variation in access. Provider availability, rural–urban differences, transportation infrastructure, and state-level treatment policies can all meaningfully shape behavioral-health service use. Prior work (e.g., Mennis & Stahler, 2015) shows that treatment density strongly affects utilization, suggesting that the present findings describe national patterns but may not generalize to local contexts. Despite these limitations, the replication of widely documented predictors (such as gender disparities, racial/ethnic gaps, the importance of insurance, and the prominence of comorbidity) strengthens confidence that the observed associations reflect persistent and meaningful mechanisms within the U.S. behavioral-health system.

Future Directions

Future research should build on these findings by extending the temporal, structural, and psychological scope of behavioral-health models. Because this study relied on cross-sectional data, longitudinal work is needed to clarify how treatment-seeking unfolds over time and to distinguish age effects from generational cohort patterns. Tracking individuals across survey waves or linking NSDUH respondents to administrative records would help establish whether predictors such as insurance, employment, or distress precede treatment entry or change as a result of receiving care.

Additional insight would come from incorporating richer attitudinal and psychological measures than those available in the public-use NSDUH file. While perceived risk offered a useful starting point, more direct assessments of stigma, perceived behavioral control, and treatment expectations would allow for stronger tests of Ajzen's Theory of Planned Behavior and help clarify the motivational pathways that lead to treatment engagement.

Finally, future research should prioritize contextual and structural factors that could not be modeled here. Linking restricted-use NSDUH data to geographic indicators, treatment facility data, or state-level policies would illuminate environmental constraints that shape behavioral-health access, particularly for low-income and minority populations. Special attention is also needed for groups not represented in NSDUH, such as incarcerated individuals, military personnel, and residents of treatment institutions, whose experiences with behavioral-health systems differ substantially from those of the general adult population.

In summary, future work should aim to clarify temporal dynamics, strengthen psychological measurement, and incorporate structural context to build a more comprehensive and causally informed understanding of behavioral-health treatment utilization.

Conclusion

Taken together, our findings show that U.S. behavioral-health treatment operates less as two separate systems and more as a shared continuum, where entry is governed by the convergence of clinical severity, structural opportunity, and psychological readiness to seek care.

Using 2023 NSDUH data and a multi-method analytic strategy that combined group LASSO, survey-weighted logistic regression, and Random Forest models, this study identified a consistent pattern in which need-based indicators and enabling resources dominated other predictors of past-year service use across both mental health and substance use treatment.

This study examined the predictors of behavioral-health treatment utilization among U.S. adults using the 2023 NSDUH, integrating insights from Andersen's Behavioral Model and Ajzen's Theory of Planned Behavior. By applying group LASSO, survey-weighted logistic regression, and Random Forest models, the analysis provided a multidimensional view of how

demographic predispositions, structural enablers, and clinical indicators jointly shape access to mental health and substance use treatment.

Across methods, several robust patterns emerged. Women consistently showed higher odds of receiving mental health treatment, whereas substance use treatment was most common among midlife adults who were likely to be navigating social, occupational, or legal consequences of substance involvement. Racial and ethnic disparities persisted across both treatment domains, highlighting longstanding inequities that were not explained by socioeconomic status or clinical need. Enabling resources, especially insurance coverage, remained central for mental health treatment, while the substance use treatment models revealed more complex pathways involving institutional referral and crisis-driven entry. Clinical need variables, particularly major depressive episodes, illicit drug use, alcohol use disorder, and marijuana involvement, were among the strongest predictors in both domains, underscoring the interconnected nature of behavioral-health conditions.

Altogether these findings suggest that mental health and substance use treatment systems function less as separate pathways and more as parallel components of a shared behavioral-health continuum. Access is shaped not only by whether individuals experience distress or substance-related problems, but also by whether they are insured, culturally supported, structurally enabled, and psychologically prepared to seek help. Reducing disparities in this space will require efforts that address both opportunity and motivation: expanding insurance and service availability while also fostering trust, reducing stigma, and improving recognition of need. Ultimately, equitable behavioral-health access depends on simultaneously strengthening the structural capacity of the system and the psychological readiness of individuals to engage with it.

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Appendix A

Survey items, table for additional information about variables

Table A1

Variable names, descriptions, data types, missingness percentages, and their roles

Variable Name	Description	Data Type	% Missingness (within sample)	Role
SUTRTPY	Past-year receipt of substance use treatment	Categorical (binary)	0%	Primary Outcome
MHTRTPY	Past-year receipt of mental health treatment	Categorical (binary)	0%	Primary Outcome
CATAG6	Age	Categorical (ordinal)	0%	Predictor (demographic)
IRSEX	Sex assigned at birth	Categorical (binary)	0%	Predictor (demographic)
NEWRACE2	Race/ethnicity	Categorical (nominal)	0%	Predictor (demographic)
IRMARIT	Marital status	Categorical (nominal)	0%	Predictor (socioeconomic)
IREDUHIGHS T2	Highest level of education completed	Categorical (ordinal)	0%	Predictor (socioeconomic)
IRWRKSTAT	Employment status	Categorical (nominal)	0%	Predictor (socioeconomic)
IRINSUR4	Health insurance coverage	Categorical (binary)	0%	Predictor (socioeconomic)
INCOME	Family income	Categorical (ordinal)	0%	Predictor (socioeconomic)
IRPYUD5ALC	Past-year alcohol use disorder	Categorical (binary)	0%	Predictor (clinical severity)

nTable A2

Variable names, descriptions, data types, missingness percentages, and their roles

Variable Name	Description	Data Type	% Missingness (within sample)	Role
IRPYUD5MRJ	Past-year marijuana use disorder	Categorical (binary)	0%	Predictor (clinical severity)
IRAMDEYR	Past-year major depressive episode	Categorical (binary)	0%	Predictor (clinical severity)
KSSLR6YR	Serious psychological distress	Numeric (count)	65.71%	Predictor (clinical severity)
IRALCFY	Frequency of alcohol use	Numeric (count)	0%	Predictor (attitudinal)
IRMJFY	Frequency of marijuana use	Numeric (count)	0%	Predictor (attitudinal)
ILLYR	Any illicit drug use in the past year	Categorical (binary)	0%	Predictor (attitudinal)
RSKCIGPKD	Risk of smoking cigarettes daily	Categorical (ordinal)	1.19%	Predictor (risk perception)
RSKMRJWK	Risk of using marijuana weekly	Categorical (ordinal)	1.42%	Predictor (risk perception)

Appendix B

Additional tables, graphs for Descriptive Statistics

Table B1

Count, percentage of unique values of categorical variables relative to total data sample size

Variable Description	<i>n</i>	%
Past-year receipt of mental health treatment		
No	33282	73.74
Yes	11851	26.26
Past-year receipt of substance use treatment		
No	42880	95.01
Yes	2253	4.99
Age		
18-25	13587	30.10
26-34	9171	20.32
35-49	12355	27.37
50-64	4960	10.99
65+	5060	11.21
Sex assigned at birth		
Male	20173	44.7
Female	24960	55.3
Race/ethnicity		
Non-Hispanic White	26054	57.73
Non-Hispanic Black/African American	5218	11.56

Table B2*Count, percentage of unique values of categorical variables relative to total data sample size*

Variable Description	<i>n</i>	%
Race/ethnicity		
Non-Hispanic Native American/Alaska Native	741	1.64
Non-Hispanic Native Hawaiian/Other Pacific Islander	223	0.49
Non-Hispanic Asian	2218	4.91
Non-Hispanic more than one race	1879	4.16
Hispanic	8800	19.50
Marital status		
Married	18309	40.57
Widowed	1376	3.05
Divorced or Separated	4664	10.33
Never Been Married	20784	46.05
Highest level of education completed		
Fifth grade or less	339	0.75
Sixth grade	254	0.56
Seventh grade	111	0.25
Eighth grade	330	0.73
Ninth grade	638	1.41
Tenth grade	776	1.72
Eleventh or Twelfth grade, no diploma	2565	5.68
High school diploma/GED	12074	26.75
Some college credit, no degree	9075	20.11
Associate's degree (AA, AS)	3954	8.76
College graduate or higher	15017	33.27

Table B3*Count, percentage of unique values of categorical variables relative to total data sample size*

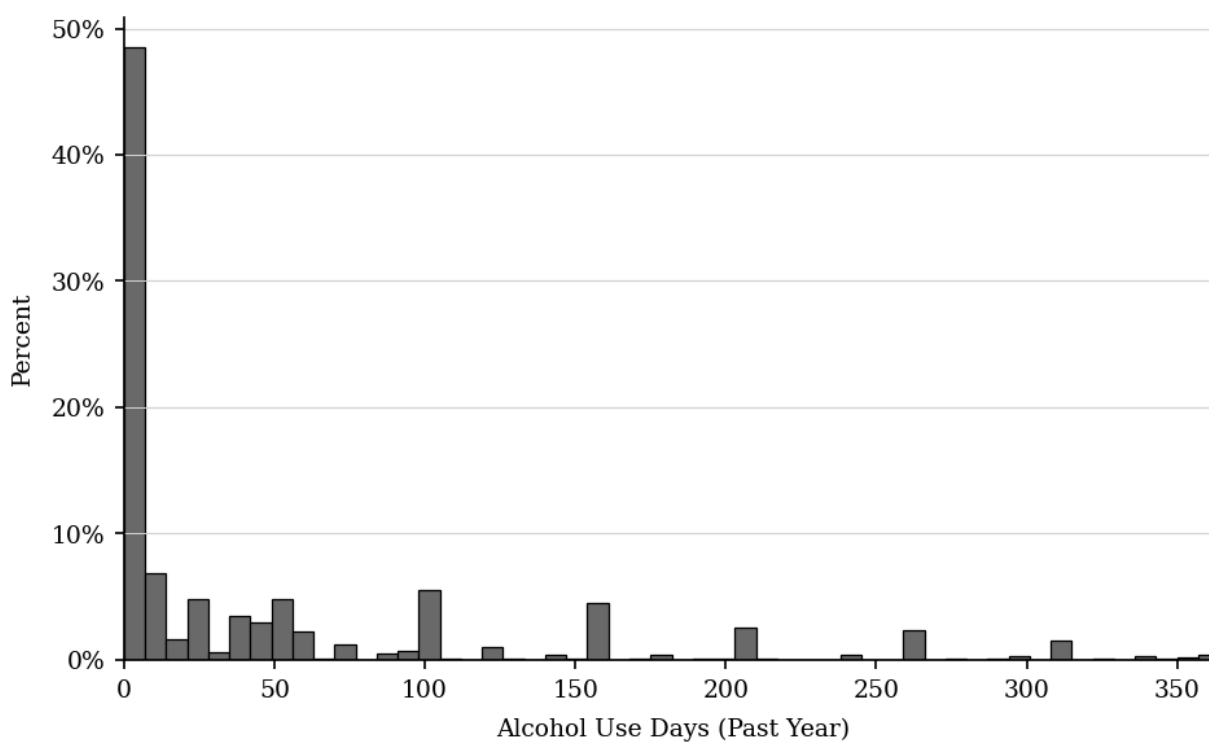
Variable Description	<i>n</i>	%
Employment status		
Employed full time	22182	49.15
Employed part time	6712	14.87
Unemployed	2795	6.19
Other (not in labor force)	13444	29.79
Health insurance coverage		
No	4816	10.67
Yes	40317	89.33
Family income		
< \$20,000	7546	16.72
\$20,000-\$49,999	12270	27.19
\$50,000-\$74,999	6626	14.68
\$75,000+	18691	41.41
Past-year alcohol use disorder		
No	39333	87.15
Yes	5800	12.85
Past-year marijuana use disorder		
No	40721	90.22
Yes	4412	9.78
Past-year major depressive episode		
No	39948	88.51
Yes	5185	11.49

Table B4*Count, percentage of unique values of categorical variables relative to total data sample size*

Variable Description	<i>n</i>	%
Any illicit drug use in the past year		
No	30995	68.67
Yes	14138	31.33
Risk of smoking cigarettes daily		
No risk	2520	5.58
Slight risk	3032	6.72
Moderate risk	9450	20.94
Great risk	29594	65.57
Risk of using marijuana weekly		
No risk	11375	25.20
Slight risk	14346	31.79
Moderate risk	9671	21.43
Great risk	9101	20.16

Figure B1

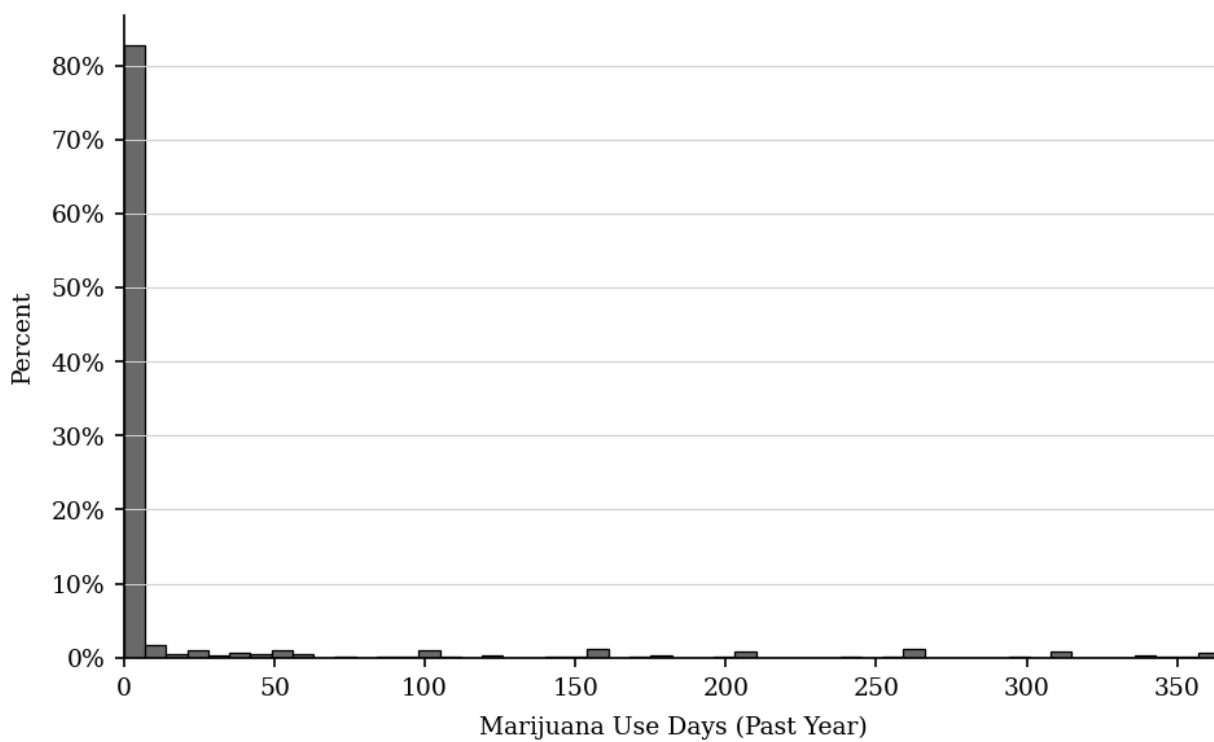
Most respondents reported few or no days of alcohol use in the past year, with a small proportion reporting frequent use.



Notes. Bars represent weighted percentages based on 2023 NSDUH person-level weights (ANALWT2_C).

Figure B2

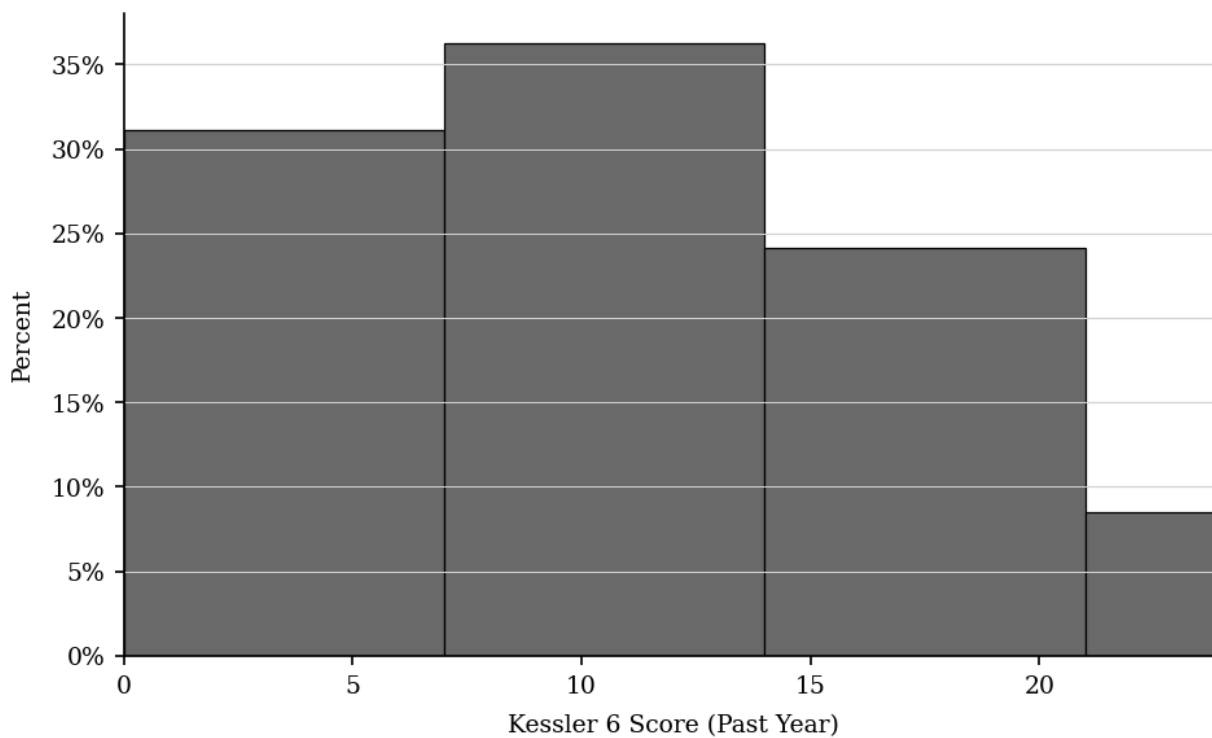
Most respondents reported no marijuana use in the past year, with few reporting frequent use.



Notes. Bars represent weighted percentages based on 2023 NSDUH person-level weights (ANALWT2_C).

Figure B3

Most respondents reported low to moderate psychological distress in the past year.



Notes. Bars represent weighted percentages based on 2023 NSDUH person-level weights (ANALWT2_C).

Appendix C

Additional tables for Group LASSO models

Table C1

Group LASSO L2 Norm of Coefficients for all variables - Mental Health Treatment Model

Variable Description	Group Size	L2 Norm	Size-Normalized L2 Norm
Age	5	0.4790	0.2142
Sex assigned at birth	2	0.4528	0.3202
Race/Ethnicity	7	0.9273	0.3505
Marital status	4	0.2173	0.1086
Highest level of education completed	11	0.7864	0.2371
Employment status	4	0.2385	0.1192
Health insurance coverage	2	0.4394	0.3107
Family income	4	0.0706	0.0353
Past-year alcohol use disorder	2	0.1734	0.1226
Past-year marijuana use disorder	2	0.2317	0.1639
Any illicit drug use in the past year	2	0.3391	0.2398
Past-year major depressive episode	2	1.2176	0.8610
Serious psychological distress	1	0.0483	0.0483
Frequency of alcohol use	1	0.0826	0.0826
Frequency of marijuana use	1	0.0692	0.0692
Risk of smoking cigarettes daily	4	0.0776	0.0388
Risk of using marijuana weekly	4	0.3470	0.1735

Table C2*Group LASSO L2 Norm of Coefficients for all variables - Substance Use Treatment Model*

Variable Description	Group Size	L2 Norm	Size-Normalized L2 Norm
Age	5	0.8739	0.3908
Sex assigned at birth	2	0.0898	0.0635
Race/Ethnicity	7	0.3937	0.1488
Marital status	4	0.3500	0.1750
Highest level of education completed	11	1.0638	0.3208
Employment status	4	0.2102	0.1051
Health insurance coverage	2	0.2425	0.1714
Family income	4	0.4687	0.2343
Past-year alcohol use disorder	2	0.7639	0.5402
Past-year marijuana use disorder	2	0.4217	0.2982
Any illicit drug use in the past year	2	0.5282	0.3735
Past-year major depressive episode	2	0.6740	0.4766
Serious psychological distress	1	0.0913	0.0913
Frequency of alcohol use	1	0.1638	0.1638
Frequency of marijuana use	1	0.0812	0.0812
Risk of smoking cigarettes daily	4	0.2186	0.1093
Risk of using marijuana weekly	4	0.1398	0.0699

Appendix D

Additional tables for weighted logistic regression models

Table D1

Logistic Regression odds ratios for selected variables - Mental Health Treatment Model

Variable Description	Odds Ratios and 95% CI (lower, upper)
Age	
18-25	n/a
26-34	1.18 (1.07, 1.30)
35-49	1.23 (1.12, 1.35)
50-64	1.03 (0.93, 1.14)
65+	0.69 (0.61, 0.77)
Sex assigned at birth	
Male	n/a
Female	1.92 (1.82, 2.02)
Race/ethnicity	
Non-Hispanic White	n/a
Non-Hispanic Black/African American	0.45 (0.41, 0.49)
Non-Hispanic Native American/Alaska Native	0.97 (0.69, 1.37)
Non-Hispanic Native Hawaiian/Other Pacific Islander	0.44 (0.28, 0.71)
Non-Hispanic Asian	0.46 (0.41, 0.52)
Non-Hispanic more than one race	0.91 (0.77, 1.08)
Hispanic	0.59 (0.55, 0.64)

Notes. Levels with n/a are reference levels being dropped during one-hot encoding.

Table D2*Logistic Regression odds ratios for selected variables - Mental Health Treatment Model*

Variable Description	Odds Ratios and 95% CI (lower, upper)
Marital status	
Married	n/a
Widowed	1.07 (0.96, 1.21)
Divorced or Separated	1.42 (1.32, 1.52)
Never Been Married	1.18 (1.10, 1.26)
Highest level of education completed	
Fifth grade or less	n/a
Sixth grade	1.47 (0.93, 2.33)
Seventh grade	1.30 (0.65, 2.59)
Eighth grade	0.71 (0.45, 1.11)
Ninth grade	1.37 (0.94, 1.98)
Tenth grade	1.32 (0.92, 1.90)
Eleventh or Twelfth grade, no diploma	0.84 (0.60, 1.17)
High school diploma/GED	0.92 (0.68, 1.25)
Some college credit, no degree	1.24 (0.91, 1.69)
Associate's degree (AA, AS)	1.25 (0.91, 1.71)
College graduate or higher	1.63 (1.20, 2.21)
Employment status	
Employed full time	n/a
Employed part time	1.33 (1.23, 1.44)
Unemployed	1.01 (0.89, 1.16)
Other (not in labor force)	1.36 (1.28, 1.45)

Notes. Levels with n/a are reference levels being dropped during one-hot encoding.

Table D3*Logistic Regression odds ratios for selected variables - Mental Health Treatment Model*

Variable Description	Odds Ratios and 95% CI (lower, upper)
Health insurance coverage	
No	n/a
Yes	1.79 (1.61, 1.98)
Past-year alcohol use disorder	
No	n/a
Yes	1.18 (1.09, 1.27)
Past-year marijuana use disorder	
No	n/a
Yes	1.19 (1.08, 1.32)
Any illicit drug use in the past year	
No	n/a
Yes	1.65 (1.55, 1.76)
Past-year major depressive episode	
No	n/a
Yes	6.17 (5.71, 6.66)
Risk of using marijuana weekly	
No risk	n/a
Slight risk	1.14 (1.07, 1.22)
Moderate risk	0.87 (0.81, 0.95)
Great risk	0.67 (0.61, 0.73)

Notes. Levels with n/a are reference levels being dropped during one-hot encoding.

Table D4*Logistic Regression odds ratios for selected variables - Substance Use Treatment Model*

Variable Description	Odds Ratios and 95% CI (lower, upper)
Age	
18-25	n/a
26-34	1.83 (1.53, 2.19)
35-49	2.68 (2.24, 3.19)
50-64	1.93 (1.59, 2.34)
65+	0.87 (0.68, 1.12)
Sex assigned at birth	
Male	n/a
Female	0.88 (0.79, 0.97)
Race/ethnicity	
Non-Hispanic White	n/a
Non-Hispanic Black/African American	0.62 (0.53, 0.73)
Non-Hispanic Native American/Alaska Native	1.40 (0.85, 2.33)
Non-Hispanic Native Hawaiian/Other Pacific Islander	1.47 (0.86, 2.49)
Non-Hispanic Asian	0.68 (0.52, 0.89)
Non-Hispanic more than one race	0.72 (0.53, 0.98)
Hispanic	0.84 (0.74, 0.96)
Marital status	
Married	n/a
Widowed	1.99 (1.58, 2.51)
Divorced or Separated	1.74 (1.52, 2.01)
Never Been Married	1.51 (1.31, 1.74)

Notes. Levels with n/a are reference levels being dropped during one-hot encoding.

Table D5*Logistic Regression odds ratios for selected variables - Substance Use Treatment Model*

Variable Description	Odds Ratios and 95% CI (lower, upper)
Highest level of education completed	
Fifth grade or less	n/a
Sixth grade	1.40 (0.54, 3.60)
Seventh grade	5.15 (2.03, 13.07)
Eighth grade	2.49 (1.16, 5.35)
Ninth grade	4.47 (2.25, 8.90)
Tenth grade	2.46 (1.21, 4.99)
Eleventh or Twelfth grade, no diploma	2.65 (1.37, 5.16)
High school diploma/GED	1.83 (0.96, 3.48)
Some college credit, no degree	1.56 (0.81, 2.98)
Associate's degree (AA, AS)	1.20 (0.62, 2.32)
College graduate or higher	1.07 (0.56, 2.05)
Employment status	
Employed full time	n/a
Employed part time	1.33 (1.14, 1.56)
Unemployed	1.40 (1.15, 1.71)
Other (not in labor force)	1.46 (1.28, 1.66)
Health insurance coverage	
No	n/a
Yes	1.71 (1.44, 2.04)

Notes. Levels with n/a are reference levels being dropped during one-hot encoding.

Table D6*Logistic Regression odds ratios for selected variables - Substance Use Treatment Model*

Variable Description	Odds Ratios and 95% CI (lower, upper)
Family income	
< \$20,000	n/a
\$20,000-\$49,999	0.69 (0.61, 0.79)
\$50,000-\$74,999	0.63 (0.53, 0.75)
\$75,000+	0.54 (0.46, 0.63)
Past-year alcohol use disorder	
No	n/a
Yes	2.32 (2.07, 2.60)
Past-year marijuana use disorder	
No	n/a
Yes	1.55 (1.35, 1.78)
Any illicit drug use in the past year	
No	n/a
Yes	2.39 (2.13, 2.68)
Past-year major depressive episode	
No	n/a
Yes	2.93 (2.60, 3.30)

Notes. Levels with n/a are reference levels being dropped during one-hot encoding.

Appendix E

Additional tables for Random Forest models

Table E1

Random Forest feature score for all variables - Mental Health Treatment Model

Variable Description	Coefficient
Age	
18-25	n/a
26-34	0.0142
35-49	0.0176
50-64	0.0157
65+	0.0132
Sex assigned at birth	
Male	n/a
Female	0.0451
Race/ethnicity	
Non-Hispanic White	n/a
Non-Hispanic Black/African American	0.0166
Non-Hispanic Native American/Alaska Native	0.0008
Non-Hispanic Native Hawaiian/Other Pacific Islander	0.0006
Non-Hispanic Asian	0.0089
Non-Hispanic more than one race	0.0033
Hispanic	0.0181

Notes. Levels with n/a are reference levels being dropped during one-hot encoding.

Table E2*Random Forest feature score for all variables - Mental Health Treatment Model*

Variable Description	Coefficient
Marital status	
Married	n/a
Widowed	0.0061
Divorced or Separated	0.0147
Never Been Married	0.0211
Highest level of education completed	
Fifth grade or less	n/a
Sixth grade	0.0007
Seventh grade	0.0001
Eighth grade	0.0004
Ninth grade	0.0018
Tenth grade	0.0026
Eleventh or Twelfth grade, no diploma	0.0051
High school diploma/GED	0.0182
Some college credit, no degree	0.0149
Associate's degree (AA, AS)	0.0098
College graduate or higher	0.0216

Notes. Levels with n/a are reference levels being dropped during one-hot encoding.

Table E3*Random Forest feature score for all variables - Mental Health Treatment Model*

Variable Description	Coefficient
Employment status	
Employed full time	n/a
Employed part time	0.0143
Unemployed	0.0047
Other (not in labor force)	0.0219
Health insurance coverage	
No	n/a
Yes	0.0092
Family income	
< \$20,000	n/a
\$20,000-\$49,999	0.0183
\$50,000-\$74,999	0.0137
\$75,000+	0.0202
Past-year alcohol use disorder	
No	n/a
Yes	0.0136
Past-year marijuana use disorder	
No	n/a
Yes	0.0093

Notes. Levels with n/a are reference levels being dropped during one-hot encoding.

Table E4*Random Forest feature score for all variables - Mental Health Treatment Model*

Variable Description	Coefficient
Any illicit drug use in the past year	
No	n/a
Yes	0.0268
Past-year major depressive episode	
No	n/a
Yes	0.1217
Serious psychological distress	0.2058
Frequency of alcohol use	0.1017
Frequency of marijuana use	0.0541
Risk of smoking cigarettes daily	
No risk	n/a
Slight risk	0.0057
Moderate risk	0.0150
Great risk	0.0179
Risk of using marijuana weekly	
No risk	n/a
Slight risk	0.0218
Moderate risk	0.0152
Great risk	0.0182

Notes. Levels with n/a are reference levels being dropped during one-hot encoding.

Table E5*Random Forest feature score for all variables - Substance Use Treatment Model*

Variable Description	Coefficient
Age	
18-25	n/a
26-34	0.0181
35-49	0.0275
50-64	0.0182
65+	0.0085
Sex assigned at birth	
Male	n/a
Female	0.0300
Race/ethnicity	
Non-Hispanic White	n/a
Non-Hispanic Black/African American	0.0110
Non-Hispanic Native American/Alaska Native	0.0038
Non-Hispanic Native Hawaiian/Other Pacific Islander	0.0022
Non-Hispanic Asian	0.0036
Non-Hispanic more than one race	0.0037
Hispanic	0.0204

Notes. Levels with n/a are reference levels being dropped during one-hot encoding.

Table E6*Random Forest feature score for all variables - Substance Use Treatment Model*

Variable Description	Coefficient
Marital status	
Married	n/a
Widowed	0.0095
Divorced or Separated	0.0227
Never Been Married	0.0262
Highest level of education completed	
Fifth grade or less	n/a
Sixth grade	0.0001
Seventh grade	0.0002
Eighth grade	0.0027
Ninth grade	0.0093
Tenth grade	0.0026
Eleventh or Twelfth grade, no diploma	0.0105
High school diploma/GED	0.0227
Some college credit, no degree	0.0187
Associate's degree (AA, AS)	0.0086
College graduate or higher	0.0164

Notes. Levels with n/a are reference levels being dropped during one-hot encoding.

Table E7*Random Forest feature score for all variables - Substance Use Treatment Model*

Variable Description	Coefficient
Employment status	
Employed full time	n/a
Employed part time	0.0134
Unemployed	0.0095
Other (not in labor force)	0.0304
Health insurance coverage	
No	n/a
Yes	0.0111
Family income	
< \$20,000	n/a
\$20,000-\$49,999	0.0246
\$50,000-\$74,999	0.0139
\$75,000+	0.0215
Past-year alcohol use disorder	
No	n/a
Yes	0.0359
Past-year marijuana use disorder	
No	n/a
Yes	0.0281

Notes. Levels with n/a are reference levels being dropped during one-hot encoding.

Table E8*Random Forest feature score for all variables - Substance Use Treatment Model*

Variable Description	Coefficient
Any illicit drug use in the past year	
No	n/a
Yes	0.0324
Past-year major depressive episode	
No	n/a
Yes	0.0433
Serious psychological distress	0.1129
Frequency of alcohol use	0.1269
Frequency of marijuana use	0.0940
Risk of smoking cigarettes daily	
No risk	n/a
Slight risk	0.0084
Moderate risk	0.0178
Great risk	0.0244
Risk of using marijuana weekly	
No risk	n/a
Slight risk	0.0282
Moderate risk	0.0139
Great risk	0.0120

Notes. Levels with n/a are reference levels being dropped during one-hot encoding.